

#1 · Short-Term Insurance and Reinsurance Coverages

What is the key structural difference between short-term insurance and long-term life insurance?

Answer

A short-term policy covers a fixed short period and can generate zero, one, or many separate loss events during that period, while a long-term life insurance policy pays a single benefit once, at death, which may occur decades after the policy was sold.

#2 · Short-Term Insurance and Reinsurance Coverages

Define per-occurrence loss (X) versus aggregate loss (S).

Answer

The per-occurrence loss X is the dollar amount of one single loss event. The aggregate loss S is the sum of all per-occurrence losses that happen to a policy or portfolio during the coverage period.

#3 · Short-Term Insurance and Reinsurance Coverages

What does the notation $X \wedge k$ mean?

Answer

$X \wedge k$ means the minimum of X and k , that is $\min(X, k)$. It equals X when X is below k and freezes at the constant k once X reaches or exceeds k .

#4 · Short-Term Insurance and Reinsurance Coverages

Give the formula for the insurer's payment under an ordinary deductible d , for a loss X .

Answer

Payment = $X - d$ if $X > d$, and Payment = 0 if $X \leq d$.
Equivalently, Payment = $X - (X \wedge d)$.

#5 · Short-Term Insurance and Reinsurance Coverages

Give the formula for the insurer's payment under a franchise deductible d , for a loss X .

Answer

Payment = 0 if $X \leq d$, and Payment = X (the full loss, not X minus d) if $X > d$.

#6 · Short-Term Insurance and Reinsurance Coverages

How does a franchise deductible differ from an ordinary deductible once the loss exceeds the deductible amount?

Answer

An ordinary deductible still subtracts d from the payment, so the payment rises smoothly from zero as X crosses d . A franchise deductible pays the entire loss once X exceeds d , creating a sudden jump in the payment at that point.

#7 · Short-Term Insurance and Reinsurance Coverages

What is the 'maximum covered loss' u in Klugman's notation, and how does it differ from the 'policy limit'?

Answer

u is the loss amount at which the policy's payment stops increasing. The maximum payment the policy will ever make (often called the policy limit in plain English) is u minus the deductible d , not u itself.

#8 · Short-Term Insurance and Reinsurance Coverages

Write the combined formula for the insurer's payment Y with deductible d , maximum covered loss u , and coinsurance α .

Answer

$Y = \alpha \times [(X \wedge u) - (X \wedge d)]$, where the loss is first capped at u , then the deductible-capped amount is subtracted, and the result is multiplied by the coinsurance factor.

#9 · Short-Term Insurance and Reinsurance Coverages

What does a coinsurance factor α represent in a short-term insurance policy?

Answer

α is the fixed percentage ($0 < \alpha \leq 1$) of the covered loss (after deductible and limit) that the insurer pays; the policyholder retains the remaining $(1 - \alpha)$ portion as shared risk.

#10 · Short-Term Insurance and Reinsurance Coverages

A policy has deductible $d = 500$, maximum covered loss $u = 25,500$, and coinsurance $\alpha = 0.75$. A loss of $X = 10,000$ occurs. What does the insurer pay?

Answer

Payment = $0.75 \times [(10,000 \wedge 25,500) - (10,000 \wedge 500)] = 0.75 \times (10,000 - 500) = 0.75 \times 9,500 = \$7,125$.

#11 · Short-Term Insurance and Reinsurance Coverages

Using the same policy ($d = 500$, $u = 25,500$, $\alpha = 0.75$), what does the insurer pay on a loss of $X = 40,000$?

Answer

Payment = $0.75 \times [(40,000 \wedge 25,500) - (40,000 \wedge 500)] = 0.75 \times (25,500 - 500) = 0.75 \times 25,000 = \$18,750$, the maximum possible payment under this coinsurance rate.

#12 · Short-Term Insurance and Reinsurance Coverages

Define the loss elimination ratio (LER) for a single loss X and deductible d .

Answer

LER = $(X \wedge d) / X$, the fraction of the loss that the deductible removes from the insurer's obligation.

#13 · Short-Term Insurance and Reinsurance Coverages

What happens to the loss elimination ratio as the deductible d increases, holding the loss distribution fixed?

Answer

The loss elimination ratio increases, since a larger deductible removes a larger dollar amount, though it rises more slowly once d already exceeds most losses in the portfolio.

#14 · Short-Term Insurance and Reinsurance Coverages

A loss of $X = 40,000$ has a deductible of $d = 500$. What is the loss elimination ratio contributed by the deductible alone?

Answer

$LER = 500 / 40,000 = 0.0125$, or 1.25%.

#15 · Short-Term Insurance and Reinsurance Coverages

How does raising a deductible affect the insurer's expected payment and the variance of that payment?

Answer

Raising the deductible reduces both the expected payment and its variance, because the insurer no longer bears the small, frequent losses that drive much of the fluctuation in claim payments.

#16 · Short-Term Insurance and Reinsurance Coverages

How does coinsurance affect the variance of the insurer's payment, compared to a deductible or limit?

Answer

Coinsurance scales the payment by a constant factor α , so it scales the variance of the payment by α squared, leaving the underlying shape of the loss distribution unchanged. A deductible or limit changes the shape as well as the scale.

#17 · Short-Term Insurance and Reinsurance Coverages

What is the purpose of reinsurance, and what is the ceding company?

Answer

Reinsurance lets an insurance company transfer part of the risk it has accepted from its own policyholders to a second company, the reinsurer. The original insurer, in this transaction, is called the ceding company.

#18 · Short-Term Insurance and Reinsurance Coverages

What is the difference between proportional and non-proportional reinsurance?

Answer

Proportional reinsurance splits every loss by a fixed percentage regardless of its size. Non-proportional reinsurance uses a dollar threshold, and the reinsurer only responds once losses cross that threshold.

#19 · Short-Term Insurance and Reinsurance Coverages

Define quota share reinsurance.

Answer

Quota share reinsurance cedes a fixed fraction c of every loss to the reinsurer, with the insurer retaining the remaining fraction $(1 - c)$, applied uniformly across the whole book of business.

#20 · Short-Term Insurance and Reinsurance Coverages

An insurer has a 40% quota share treaty and experiences a loss of \$500,000. How much does the reinsurer pay and how much does the insurer retain?

Answer

Reinsurer pays $0.40 \times 500,000 = \$200,000$. Insurer retains $0.60 \times 500,000 = \$300,000$. The two amounts sum to the original \$500,000 loss.

#21 · Short-Term Insurance and Reinsurance Coverages

Define surplus share reinsurance.

Answer

Surplus share reinsurance is proportional, but the cession percentage is set separately for each policy based on that policy's total sum insured relative to the insurer's retention limit (line), rather than being one fixed rate for the whole book.

#22 · Short-Term Insurance and Reinsurance Coverages

Under a surplus share treaty with retention line L , how is the insurer's retained fraction calculated for a policy with total sum insured F (where $F > L$)?

Answer

The insurer retains the fraction L / F of every loss on that policy, and cedes the remaining fraction $(F - L) / F$, called the surplus, to the reinsurer.

#23 · Short-Term Insurance and Reinsurance Coverages

What happens under a surplus share treaty if a policy's total sum insured F is less than or equal to the retention line L ?

Answer

The insurer keeps the entire policy and cedes nothing to the reinsurer under the surplus treaty, since the policy does not exceed the retention line.

#24 · Short-Term Insurance and Reinsurance Coverages

An insurer sets a retention line of \$100,000. A policy has a total sum insured of \$400,000. What is the insurer's retained fraction for this policy?

Answer

Retained fraction = $100,000 / 400,000 = 0.25$ (25%).
The ceded fraction is $300,000 / 400,000 = 0.75$ (75%).

#25 · Short-Term Insurance and Reinsurance Coverages

Using the surplus share example (retention line \$100,000, sum insured \$400,000, retained fraction 25%), a loss of \$120,000 occurs on this policy. What does the insurer retain and what does the reinsurer pay?

Answer

Insurer retains $0.25 \times 120,000 = \$30,000$. Reinsurer pays $0.75 \times 120,000 = \$90,000$. The two amounts sum to the \$120,000 loss.

#26 · Short-Term Insurance and Reinsurance Coverages

What is per-occurrence excess-of-loss reinsurance, and how are the retention M and layer limit L defined?

Answer

It is non-proportional reinsurance applied separately to each loss event. The insurer sets a retention (attachment point) M, and the reinsurer's layer covers losses from M up to $M + L$, where L is the width of the layer.

#27 · Short-Term Insurance and Reinsurance Coverages

Give the formula for the amount ceded to the reinsurer under per-occurrence excess-of-loss reinsurance with retention M and layer limit L , for a loss X .

Answer

Ceded = $\min(\max(X - M, 0), L)$. The insurer retains the loss minus this ceded amount.

#28 · Short-Term Insurance and Reinsurance Coverages

Under excess-of-loss reinsurance with retention $M = 200,000$ and layer limit $L = 800,000$, a loss of $X = 650,000$ occurs. How much is ceded and how much is retained?

Answer

Ceded = $\min(\max(650,000 - 200,000, 0), 800,000)$
= $\min(450,000, 800,000) = \$450,000$. Retained =
 $650,000 - 450,000 = \$200,000$.

#29 · Short-Term Insurance and Reinsurance Coverages

Under the same excess-of-loss treaty ($M = 200,000$, $L = 800,000$), a much larger loss of $X = 1,300,000$ occurs. How much is ceded and how much is retained?

Answer

Ceded = $\min(\max(1,300,000 - 200,000, 0), 800,000) = \min(1,100,000, 800,000) = \$800,000$ (the full layer). Retained = $1,300,000 - 800,000 = \$500,000$.

#30 · Short-Term Insurance and Reinsurance Coverages

Why does the insurer's retained loss increase again once a loss exceeds the top of the excess-of-loss reinsurance layer ($M + L$)?

Answer

Because the reinsurance layer is exhausted once the loss exceeds $M + L$; the reinsurer's payment is capped at L , so any loss amount above $M + L$ reverts to being the insurer's own responsibility.

#31 · Short-Term Insurance and Reinsurance Coverages

What is aggregate excess-of-loss (stop-loss) reinsurance, and how does it differ from per-occurrence excess-of-loss reinsurance?

Answer

Stop-loss reinsurance applies the same excess-of-loss formula, but to the aggregate loss S for the whole portfolio over the period, rather than to each individual loss separately, protecting against many moderate losses accumulating into an unexpectedly large total.

#32 · Short-Term Insurance and Reinsurance Coverages

Give the formula for the reinsurer's payment under stop-loss reinsurance with aggregate retention M and aggregate limit L , applied to aggregate loss S .

Answer

Reinsurer's payment = $\min(\max(S - M, 0), L)$. The insurer keeps the rest of the aggregate loss.

#33 · Short-Term Insurance and Reinsurance Coverages

What arithmetic identity should always hold true between gross loss, ceded loss, and retained loss under any reinsurance structure?

Answer

Ceded loss plus retained loss must always equal the gross loss exactly; this is a useful check on any reinsurance calculation, since a mistake in either figure usually shows up as a mismatch when the two are added back together.

#34 · Short-Term Insurance and Reinsurance Coverages

Why is excess-of-loss reinsurance considered better than quota share at controlling an insurer's worst-case exposure to a single catastrophic loss?

Answer

Excess-of-loss reinsurance leaves the insurer fully exposed only up to the retention M and transfers everything above that point (up to the layer limit), directly targeting the extreme tail, while quota share only ever removes a fixed fraction, leaving the insurer still exposed to a fixed percentage of even the largest losses.

#35 · Short-Term Insurance and Reinsurance Coverages

Why is quota share considered better than excess-of-loss reinsurance at reducing an insurer's overall volume of risk?

Answer

Quota share removes a fixed percentage from every loss, big or small, so it reduces exposure uniformly across the whole distribution, including the ordinary, everyday losses that excess-of-loss reinsurance does not touch at all since they typically fall below the retention.

#36 · Short-Term Insurance and Reinsurance Coverages

When a policy question combines a deductible, limit, and coinsurance with a reinsurance treaty, in what order should the two layers be applied?

Answer

First calculate what the direct insurance policy pays the policyholder using $Y = \alpha \times [(X \wedge u) - (X \wedge d)]$. Then apply the reinsurance structure to that already-calculated insurer payment, since the insurer is reinsuring its own payment, not the raw underlying loss.

#37 · Short-Term Insurance and Reinsurance Coverages

A loss of $X = 300$ occurs on a policy with deductible $d = 500$. What does the insurer pay, and why?

Answer

The insurer pays \$0, because the loss is below the deductible, so both $X \wedge u$ and $X \wedge d$ equal 300, and their difference is zero before coinsurance is even applied.

#38 · Short-Term Insurance and Reinsurance Coverages

In what fixed order should the deductible-limit-coinsurance formula be evaluated to avoid common arithmetic mistakes?

Answer

First cap the loss at the maximum covered loss u , then subtract the loss capped at the deductible d , and only then multiply the result by the coinsurance factor α .

#39 · Short-Term Insurance and Reinsurance Coverages

What clue in a reinsurance question signals that proportional reinsurance (rather than excess-of-loss) is being described?

Answer

A fixed percentage applied to every loss regardless of size signals proportional reinsurance (quota share or surplus share); a dollar threshold plus a layer width signals excess-of-loss reinsurance instead.

#40 · Short-Term Insurance and Reinsurance Coverages

Why does a fixed-dollar deductible eliminate a shrinking percentage of larger and larger losses?

Answer

Because the loss elimination ratio for the deductible alone is d divided by X , and as X grows while d stays fixed, that ratio (d/X) gets smaller, so the deductible accounts for an ever-smaller share of bigger losses.

#41 · Severity Frequency and Aggregate Models

In the collective risk model, what is the formula for aggregate loss S ?

Answer

$S = X_1 + X_2 + \dots + X_N$, where N is the frequency (claim count) and each X_i is an iid severity. If $N = 0$, then $S = 0$.

#42 · Severity Frequency and Aggregate Models

What does the term 'iid' mean when describing the severities $X_1, X_2, \dots$ in the collective risk model?

Answer

Independent and identically distributed: each X_i is independent of the others and all X_i follow the same severity distribution, and N is independent of every X_i .

#43 · Severity Frequency and Aggregate Models

What is the mean and variance of the Poisson(λ) frequency distribution?

Answer

$E[N] = \lambda$ and $\text{Var}(N) = \lambda$. Mean equals variance, which is why the Poisson has no overdispersion or underdispersion.

#44 · Severity Frequency and Aggregate Models

What is the probability function of the Poisson(λ) distribution?

Answer

$p_k = P(N = k) = (e^{-\lambda} \lambda^k) / k!$ for $k = 0, 1, 2, \dots$

#45 · Severity Frequency and Aggregate Models

What is the mean and variance of the binomial(m, q) frequency distribution?

Answer

$E[N] = mq$ and $\text{Var}(N) = mq(1-q)$. Since $1-q < 1$, the variance is always less than the mean, called underdispersion.

#46 · Severity Frequency and Aggregate Models

What is the mean and variance of the negative binomial(r, β) frequency distribution?

Answer

$E[N] = r\beta$ and $\text{Var}(N) = r\beta(1+\beta)$. Since $(1+\beta) > 1$, the variance always exceeds the mean, called overdispersion.

#47 · Severity Frequency and Aggregate Models

What is the mean and variance of the geometric(β) distribution?

Answer

$E[N] = \beta$ and $\text{Var}(N) = \beta(1+\beta)$. The geometric is the special case of the negative binomial with $r = 1$.

#48 · Severity Frequency and Aggregate Models

What is the (a,b,0) class recursion?

Answer

$p_k / p_{k-1} = a + b/k$ for $k = 1, 2, 3, \dots$, where a and b are constants determined by the distribution and its parameters, and p_0 is given directly.

#49 · Severity Frequency and Aggregate Models

What are the $(a,b,0)$ parameters a and b for the $\text{Poisson}(\lambda)$ distribution?

Answer

$a = 0$ and $b = \lambda$, with $p_0 = e^{-\lambda}$.

#50 · Severity Frequency and Aggregate Models

What are the $(a,b,0)$ parameters a and b for the $\text{binomial}(m,q)$ distribution?

Answer

$a = -q/(1-q)$ and $b = (m+1)q/(1-q)$, with $p_0 = (1-q)^m$.

#51 · Severity Frequency and Aggregate Models

What are the $(a,b,0)$ parameters a and b for the negative binomial (r,β) distribution?

Answer

$a = \beta/(1+\beta)$ and $b = (r-1)\beta/(1+\beta)$, with $p_0 = (1+\beta)^{-r}$.

#52 · Severity Frequency and Aggregate Models

What are the $(a,b,0)$ parameters a and b for the geometric (β) distribution?

Answer

$a = \beta/(1+\beta)$ and $b = 0$, with $p_0 = 1/(1+\beta)$.

#53 · Severity Frequency and Aggregate Models

How can you tell from the sign of a in the $(a,b,0)$ class whether a frequency distribution is overdispersed or underdispersed?

Answer

$a > 0$ signals overdispersion (negative binomial, geometric); $a < 0$ signals underdispersion (binomial); $a = 0$ signals mean equals variance (Poisson).

#54 · Severity Frequency and Aggregate Models

What is the mean and variance of the exponential(θ) severity distribution?

Answer

$E[X] = \theta$ and $\text{Var}(X) = \theta^2$. θ is both the single parameter and the mean.

#55 · Severity Frequency and Aggregate Models

What is the density function of the exponential(θ) distribution?

Answer

$f(x) = (1/\theta) e^{-x/\theta}$ for $x > 0$.

#56 · Severity Frequency and Aggregate Models

What is the memoryless property of the exponential distribution?

Answer

Given a loss has already exceeded amount d , the amount by which it exceeds d still follows the same exponential distribution with the same mean θ .

#57 · Severity Frequency and Aggregate Models

What is the mean and variance of the gamma(α, θ) severity distribution?

Answer

$E[X] = \alpha\theta$ and $\text{Var}(X) = \alpha\theta^2$, where α is the shape parameter and θ is the scale parameter.

#58 · Severity Frequency and Aggregate Models

How is the exponential distribution a special case of the gamma distribution?

Answer

When $\alpha = 1$, the gamma(α, θ) reduces to the exponential(θ).

#59 · Severity Frequency and Aggregate Models

What is the mean of the two-parameter Pareto(α, θ) distribution, and when does it exist?

Answer

$E[X] = \theta/(\alpha-1)$, which exists only when $\alpha > 1$.

#60 · Severity Frequency and Aggregate Models

What is the variance of the two-parameter Pareto(α, θ) distribution, and when does it exist?

Answer

$\text{Var}(X) = \alpha\theta^2 / [(\alpha-1)^2 (\alpha-2)]$, which exists only when $\alpha > 2$.

#61 · Severity Frequency and Aggregate Models

Why is the Pareto distribution the standard choice for liability and catastrophe severity modeling?

Answer

The Pareto has a heavier right tail than the exponential or gamma, assigning meaningfully more probability to very large claims.

#62 · Severity Frequency and Aggregate Models

How are the parameters μ and σ of a $\text{lognormal}(\mu, \sigma)$ distribution defined?

Answer

μ and σ are the mean and standard deviation of $\ln(X)$, the natural log of X , not of X itself.

#63 · Severity Frequency and Aggregate Models

What is the mean of the lognormal(μ, σ) distribution?

Answer

$$E[X] = e^{\{\mu + \sigma^2/2\}}.$$

#64 · Severity Frequency and Aggregate Models

What is the variance of the lognormal(μ, σ) distribution?

Answer

$$\text{Var}(X) = e^{\{2\mu + \sigma^2\}} (e^{\{\sigma^2\}} - 1).$$

#65 · Severity Frequency and Aggregate Models

What is the mean of the Weibull(θ, τ) distribution?

Answer

$E[X] = \theta \Gamma(1 + 1/\tau)$, where Γ is the gamma function, a continuous extension of the factorial function.

#66 · Severity Frequency and Aggregate Models

What is the variance of the Weibull(θ, τ) distribution?

Answer

$\text{Var}(X) = \theta^2 [\Gamma(1 + 2/\tau) - \Gamma(1 + 1/\tau)^2]$.

#67 · Severity Frequency and Aggregate Models

How is the exponential distribution a special case of the Weibull distribution?

Answer

When $\tau = 1$, the Weibull(θ, τ) reduces to the exponential(θ).

#68 · Severity Frequency and Aggregate Models

How does an ordinary deductible d change the frequency of reported claims?

Answer

Only losses exceeding d are ever reported, so if $v = P(X > d)$, the reported frequency N^L is a thinned version of N : when N is $(a, b, 0)$ with parameters a and b , N^L is $(a, b, 0)$ with the same a but b replaced by $v \times b$, and p_0 recalculated.

#69 · Severity Frequency and Aggregate Models

If the true underlying claim frequency is $\text{Poisson}(\lambda)$ and a deductible causes only a fraction v of losses to be reported, what is the distribution of the reported frequency?

Answer

$\text{Poisson}(v\lambda)$. Thinning a Poisson frequency by a constant probability v produces another Poisson distribution with mean scaled by v .

#70 · Severity Frequency and Aggregate Models

What is the left-truncated and shifted severity distribution Y^P , and what does it represent?

Answer

$Y^P = X - d$, given $X > d$. It represents the amount actually paid per payment, conditional on a payment being made at all.

#71 · Severity Frequency and Aggregate Models

What is the formula for $E[Y^P]$, the expected payment per payment given a deductible d ?

Answer

$E[Y^P] = E[(X-d)_+] / [1 - F_X(d)]$, where $1 - F_X(d)$ is the survival function $S_X(d)$, the probability a loss exceeds d .

#72 · Severity Frequency and Aggregate Models

What is $E[(X-d)_+]$, and how does it differ from $E[Y^P]$?

Answer

$E[(X-d)_+]$ is the expected payment per loss, averaged across every loss including those below the deductible that pay nothing. $E[Y^P]$ is the expected payment per payment, conditional only on losses that exceed d .

#73 · Severity Frequency and Aggregate Models

What is the formula relating $E[(X-d)_+]$ to the limited expected value $E[X \wedge d]$?

Answer

$E[(X-d)_+] = E[X] - E[X \wedge d]$, where $X \wedge d$ is the smaller of X and d .

#74 · Severity Frequency and Aggregate Models

If severity X is exponential with mean θ and a deductible d is applied, what is $E[Y^P]$, the expected payment per payment?

Answer

$E[Y^P] = \theta$, regardless of the size of d . This follows directly from the exponential's memoryless property.

#75 · Severity Frequency and Aggregate Models

What is a policy limit u , and what does it produce mathematically?

Answer

A policy limit u caps the maximum amount an insurer pays on a single claim. It produces a right-censored payment amount $Y = X \wedge u$: the full loss is paid below u , and exactly u is paid at or above u .

#76 · Severity Frequency and Aggregate Models

How does right-censoring from a policy limit differ from left-truncation from a deductible?

Answer

A deductible removes information about small losses (never reported at all). A policy limit caps information about large losses (the insurer never learns exactly how large a loss above u was, only that it was at least u).

#77 · Severity Frequency and Aggregate Models

With both a deductible d and a limit u , what is the insurer's payment on a given loss X ?

Answer

$(X \wedge u) - (X \wedge d)$: zero for losses below d , rising dollar for dollar between d and u , and capped at $u - d$ for losses at or above u .

#78 · Severity Frequency and Aggregate Models

What does coinsurance α do to a payment, and in what order should it be applied relative to a deductible and limit?

Answer

Coinsurance α proportionally scales the payment: the amount becomes $\alpha \times [(X \wedge u) - (X \wedge d)]$. It should always be applied after the deductible and limit have already determined the covered loss.

#79 · Severity Frequency and Aggregate Models

How does coinsurance α affect the mean, standard deviation, and variance of a payment?

Answer

It multiplies the mean and standard deviation by α , and multiplies the variance by α^2 , since coinsurance is a pure multiplicative scaling.

#80 · Severity Frequency and Aggregate Models

What is the formula for the mean of the aggregate loss S in the collective risk model?

Answer

$E[S] = E[N] \times E[X]$, the expected number of claims times the expected size of each claim.

#81 · Severity Frequency and Aggregate Models

What is the formula for the variance of the aggregate loss S in the collective risk model?

Answer

$$\text{Var}(S) = E[N] \times \text{Var}(X) + \text{Var}(N) \times E[X]^2.$$

#82 · Severity Frequency and Aggregate Models

In the aggregate loss variance formula $\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2$, what does the first term represent?

Answer

The variability in S caused by claim sizes varying around their mean, weighted by how many claims there typically are.

#83 · Severity Frequency and Aggregate Models

In the aggregate loss variance formula $\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2$, what does the second term represent?

Answer

The variability in S caused by the claim count itself varying, weighted by the square of the average claim size.

#84 · Severity Frequency and Aggregate Models

Why can the variance of an aggregate loss be much larger relative to its mean when frequency is negative binomial rather than Poisson?

Answer

The negative binomial is overdispersed ($\text{Var}(N) > E[N]$), so the $\text{Var}(N) \times E[X]^2$ term in the aggregate variance formula is inflated compared to a Poisson frequency with the same mean.

#85 · Severity Frequency and Aggregate Models

If N is $\text{Poisson}(\lambda)$ and severity is X , what shortcut formula gives $\text{Var}(S)$ directly?

Answer

$\text{Var}(S) = \lambda \times E[X^2]$, since for a Poisson frequency $E[N] = \text{Var}(N) = \lambda$.

#86 · Severity Frequency and Aggregate Models

Worked example: frequency N is $\text{Poisson}(\lambda = 4)$ and severity X is exponential with $\theta = 500$. What is $E[S]$?

Answer

$E[S] = E[N] \times E[X] = 4 \times 500 = 2,000$.

#87 · Severity Frequency and Aggregate Models

Worked example: frequency N is Poisson($\lambda = 4$) and severity X is exponential with $\theta = 500$ (so $\text{Var}(X) = \theta^2 = 250,000$). What is $\text{Var}(S)$?

Answer

$$\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2 = 4 \times 250,000 + 4 \times 500^2 = 1,000,000 + 1,000,000 = 2,000,000.$$

#88 · Severity Frequency and Aggregate Models

Worked example: N is negative binomial with $r = 2$, $\beta = 3$. What are $E[N]$ and $\text{Var}(N)$?

Answer

$$E[N] = r\beta = 2 \times 3 = 6. \quad \text{Var}(N) = r\beta(1+\beta) = 2 \times 3 \times 4 = 24.$$

#89 · Severity Frequency and Aggregate Models

Worked example: severity X is gamma with $\alpha = 2$, $\theta = 100$. What are $E[X]$ and $\text{Var}(X)$?

Answer

$$E[X] = \alpha\theta = 2 \times 100 = 200. \text{Var}(X) = \alpha\theta^2 = 2 \times 100^2 = 20,000.$$

#90 · Severity Frequency and Aggregate Models

Worked example: with N negative binomial ($E[N]=6$, $\text{Var}(N)=24$) and X gamma ($E[X]=200$, $\text{Var}(X)=20,000$), what is $E[S]$?

Answer

$$E[S] = E[N] \times E[X] = 6 \times 200 = 1,200.$$

#91 · Severity Frequency and Aggregate Models

Worked example: with N negative binomial ($E[N]=6$, $\text{Var}(N)=24$) and X gamma ($E[X]=200$, $\text{Var}(X)=20,000$), what is $\text{Var}(S)$?

Answer

$$\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2 = 6 \times 20,000 + 24 \times 200^2 = 120,000 + 960,000 = 1,080,000.$$

#92 · Severity Frequency and Aggregate Models

What conditions must hold for the Panjer recursion to compute the full distribution of S?

Answer

Frequency N must belong to the (a,b,0) class, and severity X must be discrete (or discretized onto a grid of equally spaced values).

#93 · Severity Frequency and Aggregate Models

What is the Panjer recursion formula for $f_S(k)$ when $k \geq 1$?

Answer

$f_S(k) = [1 / (1 - a f_X(0))] \times \sum_{j=1}^k (a + bj/k) f_X(j) f_S(k-j)$, using the same a and b as the frequency's $(a,b,0)$ class.

#94 · Severity Frequency and Aggregate Models

Why is the Panjer recursion computationally efficient compared to a full convolution of N and X ?

Answer

It reuses the $(a,b,0)$ recursive structure of the frequency distribution to build $f_S(k)$ from previously computed, smaller values of f_S , rather than requiring a full convolution.

#95 · Severity Frequency and Aggregate Models

Worked example (Panjer): N is $\text{Poisson}(\lambda=2)$, so $a=0$ and $b=2$. What is $f_S(0)$?

Answer

$f_S(0) = P(N=0) = e^{-\lambda} = e^{-2} \approx 0.135335$, since for the $(a,b,0)$ class p_0 is given directly.

#96 · Severity Frequency and Aggregate Models

Worked example (Panjer): N is $\text{Poisson}(\lambda=2)$ with $a=0$, severity $f_X(1)=0.6$, $f_X(2)=0.4$. What is the simplified recursion formula when $a=0$?

Answer

$f_S(k) = (2/k) \times \sum_{j=1}^k j \times f_X(j) \times f_S(k-j)$, since $a=0$ removes the a -term entirely.

#97 · Severity Frequency and Aggregate Models

Worked example (Panjer): with $f_S(0) \approx 0.135335$, $f_X(1)=0.6$, what is $f_S(1)$?

Answer

$$f_S(1) = (2/1) \times [1 \times f_X(1) \times f_S(0)] = 2 \times 0.6 \times 0.135335 \approx 0.162402.$$

#98 · Severity Frequency and Aggregate Models

Worked example (Panjer): with $f_S(0) \approx 0.135335$, $f_S(1) \approx 0.162402$, $f_X(1)=0.6$, $f_X(2)=0.4$, what is $f_S(2)$?

Answer

$$\begin{aligned} f_S(2) &= (2/2) \times [1 \times f_X(1) \times f_S(1) + 2 \times f_X(2) \times f_S(0)] \\ &= 0.6 \times 0.162402 + 2 \times 0.4 \times 0.135335 \approx 0.097441 + 0.108268 = 0.205709. \end{aligned}$$

#99 · Severity Frequency and Aggregate Models

How can a Panjer recursion result for $f_S(2)$ be cross-checked directly for a Poisson($\lambda=2$) frequency with two-point severity $f_X(1)=0.6$, $f_X(2)=0.4$?

Answer

By adding the two ways $S=2$ can occur: $P(N=1) \times f_X(2) + P(N=2) \times f_X(1)^2$, using $P(N=1)=P(N=2)=2e^{-2} \approx 0.270671$, giving $0.270671 \times 0.4 + 0.270671 \times 0.36 \approx 0.205710$.

#100 · Severity Frequency and Aggregate Models

What is a 'loss' versus a 'claim' or 'payment' in the collective risk model?

Answer

A loss is any event that could trigger a payment. A claim or payment is the amount actually paid after any deductible, limit, or coinsurance provision has been applied.

#101 · Severity Frequency and Aggregate Models

What does the survival function $S_X(d)$ represent in the context of a deductible d ?

Answer

$S_X(d) = 1 - F_X(d) = P(X > d)$, the probability a loss exceeds the deductible, also written v .

#102 · Severity Frequency and Aggregate Models

What is the limited expected value $E[X \wedge d]$?

Answer

The expected value of X capped at d : $E[X \wedge d]$ represents the average payment if every loss were subject to a maximum payout of d with no deductible.

#103 · Severity Frequency and Aggregate Models

Why is it common practice to discretize a continuous severity distribution before applying the Panjer recursion?

Answer

The Panjer recursion requires a discrete severity probability function $f_X(j)$; continuous severities are placed onto a grid of equally spaced values so the recursion can be applied in practice.

#104 · Severity Frequency and Aggregate Models

What real-world reason makes the negative binomial a better frequency model than the Poisson for a heterogeneous population?

Answer

When some policyholders are riskier than others in ways the model cannot observe, claim counts are more volatile than a Poisson (mean = variance) would predict; the negative binomial's overdispersion (variance > mean) captures this.

#105 · Severity Frequency and Aggregate Models

In the $(a, b, 0)$ class recursion $p_k/p_{k-1} = a + b/k$, what does the '0' in the class name refer to?

Answer

It signals that the recursion starts at $k = 0$, with p_0 given directly rather than derived from the $a + b/k$ formula.

#106 · Severity Frequency and Aggregate Models

What is the correct order of operations when a policy has a deductible, a limit, and coinsurance all at once?

Answer

Apply the deductible first, then the limit, then coinsurance last: $\text{payment} = \alpha \times [(X \wedge u) - (X \wedge d)]$.

#107 · Severity Frequency and Aggregate Models

Why does N need to be independent of the X_i values in the collective risk model?

Answer

So that how many claims occur does not change the size of any individual claim, which is required for the $E[S]$ and $\text{Var}(S)$ formulas to hold.

#108 · Severity Frequency and Aggregate Models

What happens to the reported claim frequency's $(a, b, 0)$ parameters a and b when a deductible thins the original frequency N ?

Answer

The parameter a stays the same, but b is replaced by $v \times b$ (where $v = P(X > d)$), and p_0 is recalculated for the thinned distribution.

#109 · Severity Frequency and Aggregate Models

What is the shape parameter role of α in the gamma distribution, versus its role in coinsurance?

Answer

In the gamma (and Pareto) distribution, α is the shape parameter controlling the distribution's form. In coinsurance, α is an unrelated symbol representing the insurer's payment percentage; the two uses must not be confused.

#110 · Severity Frequency and Aggregate Models

Which severity distribution is the standard choice for modeling normal, moderate-sized losses using techniques borrowed from the normal distribution?

Answer

The lognormal distribution, because $\ln(X)$ is normally distributed, so normal distribution tables and calculations can be reused directly.

#111 · Severity Frequency and Aggregate Models

What is $E[X^2]$ in terms of $E[X]$ and $\text{Var}(X)$, and why is it useful for a Poisson aggregate loss calculation?

Answer

$E[X^2] = \text{Var}(X) + E[X]^2$. It is useful because for Poisson frequency, $\text{Var}(S) = \lambda \times E[X^2]$ is a direct shortcut avoiding separate computation of the two variance terms.

#112 · Severity Frequency and Aggregate Models

Why does an insurer's payment under a deductible-only policy equal $E[X] - E[X \wedge d]$ on a per-loss basis?

Answer

$E[X \wedge d]$ represents what the insurer would have paid if payments were capped at d with no deductible; subtracting it from the true mean $E[X]$ isolates exactly the portion of loss exceeding d , which is $E[(X-d)_+]$.

#113 · Severity Frequency and Aggregate Models

If a Poisson frequency N has $\lambda = 4$ and severity X is exponential with $\theta = 500$, what is $\text{Var}(X)$?

Answer

$\text{Var}(X) = \theta^2 = 500^2 = 250,000$, since for the exponential distribution variance equals the mean squared.

#114 · Severity Frequency and Aggregate Models

What is the standard deviation of S when $\text{Var}(S) = 2,000,000$?

Answer

$\text{SD}(S) = \sqrt{2,000,000} \approx 1,414.21$.

#115 · Severity Frequency and Aggregate Models

Which $(a,b,0)$ frequency distribution has $b = 0$, and what does that mean for its recursion?

Answer

The geometric distribution has $b = 0$, so its recursion simplifies to $p_k/p_{k-1} = a$, meaning each probability is a constant multiple of the one before it.

#116 · Parametric Estimation

What is the method of moments?

Answer

It is an estimation technique that sets sample moments (like the sample mean and sample second moment) equal to the corresponding theoretical moments of the distribution, then solves the resulting equations for the unknown parameters.

#117 · Parametric Estimation

For a sample of n observations, what is the first sample moment m_1 ?

Answer

$m_1 = (1/n)$ times the sum of the observations, which is simply the sample mean, $\bar{x}$.

#118 · Parametric Estimation

For a gamma distribution with shape α and scale θ , what two equations does the method of moments use?

Answer

Set α times θ equal to the sample mean, and α times θ^2 equal to the sample variance, then solve the two equations together for α and θ .

#119 · Parametric Estimation

What is a key weakness of the method of moments compared to maximum likelihood?

Answer

It only uses a couple of summary numbers from the sample, so it wastes information, and it has no natural way to incorporate censored or truncated data.

#120 · Parametric Estimation

What is the likelihood function $L(\theta)$?

Answer

It is the joint probability, or joint density, of observing the exact sample that was collected, written as a function of the unknown parameter θ , equal to the product of each observation's individual density.

#121 · Parametric Estimation

Why do actuaries work with the log-likelihood instead of the likelihood itself?

Answer

Taking logs turns a product of many terms into a sum, which is far easier to differentiate, and since the natural logarithm is increasing, the value of θ that maximizes the log-likelihood also maximizes the likelihood.

#122 · Parametric Estimation

What is the likelihood equation?

Answer

It is the equation formed by setting the derivative of the log-likelihood with respect to θ equal to zero, written as the partial derivative of $l(\theta)$ with respect to θ equals zero; its solution is the maximum likelihood estimate.

#123 · Parametric Estimation

For an exponential distribution with mean θ , what is the MLE of θ given a complete sample?

Answer

The MLE of θ always equals the sample mean, the sum of the observations divided by the sample size n , which is also the method of moments estimator for this particular distribution.

#124 · Parametric Estimation

How does the likelihood function change for grouped data reported in intervals?

Answer

Each interval contributes the probability that a claim falls in that interval, found from the difference of the cumulative distribution function at the two interval endpoints, raised to the power of the number of observations in that interval.

#125 · Parametric Estimation

What is right-censored data, and how does it change the likelihood contribution?

Answer

Right-censored data occurs when a loss is capped by a policy limit u , so the true amount above u is unknown; instead of a density term, the observation contributes the survival function $S(u; \theta)$, the probability the true loss was at least u .

#126 · Parametric Estimation

What is left-truncated data, and how does it change the likelihood contribution?

Answer

Left-truncated data occurs when a deductible d means losses below d are never observed; each observed loss contributes the conditional density $f(x; \theta) / S(d; \theta)$, rather than the plain density $f(x; \theta)$.

#127 · Parametric Estimation

In a small example, an exponential book has one complete claim of 4 and one claim censored at a policy limit of 10. What is the MLE of the mean θ ?

Answer

The likelihood is $(1/\theta)$ times e to the negative 14 over θ , and solving the likelihood equation gives $\hat{\theta}$ equal to 14, which is much larger than the $\hat{\theta}$ of 4 you would get by wrongly ignoring the censored claim.

#128 · Parametric Estimation

What special property of the exponential distribution simplifies left-truncated likelihood calculations?

Answer

The memoryless property means the ratio of the survival function at x plus d to the survival function at d equals the survival function at x alone, so a left-truncated exponential observation behaves like a complete observation of the excess amount over the deductible.

#129 · Parametric Estimation

How is the bias of an estimator θ -hat defined?

Answer

Bias(θ -hat) equals $E[\theta$ -hat] minus θ , the true parameter, measuring how far the estimator's average value over repeated samples sits from the truth.

#130 · Parametric Estimation

What does it mean for an estimator to be unbiased?

Answer

It means Bias(θ -hat) equals zero for every possible true value of θ , so the estimator neither systematically overestimates nor underestimates the parameter.

#131 · Parametric Estimation

What does it mean for an estimator to be consistent?

Answer

It means the estimator converges in probability to the true parameter value as the sample size n approaches infinity, so with enough data the estimate is almost certain to land close to the truth.

#132 · Parametric Estimation

Can a biased estimator still be consistent?

Answer

Yes; an estimator can have nonzero bias in small samples yet still be consistent, as long as that bias shrinks toward zero as the sample size grows without bound.

#133 · Parametric Estimation

What is the formula for mean squared error, and what does it measure?

Answer

MSE($\hat{\theta}$) equals $\text{Var}(\hat{\theta})$ plus $\text{Bias}(\hat{\theta})^2$, combining an estimator's variability and its systematic error into a single number used to compare estimators.

#134 · Parametric Estimation

Two estimators have the same MSE for a given true parameter value. What does this tell you about their bias and variance?

Answer

It only tells you the sum of their variance and squared bias is equal; one estimator could be unbiased with high variance while the other is slightly biased with low variance, and MSE alone cannot distinguish the two patterns.

#135 · Parametric Estimation

What is the asymptotic normality property of the MLE?

Answer

As the sample size n grows large, the sampling distribution of the MLE $\hat{\theta}$ becomes approximately normal, centered at the true parameter θ , with a variance that can be estimated directly from the sample.

#136 · Parametric Estimation

What is the expected (Fisher) information, $I(\theta)$?

Answer

$I(\theta)$ equals the negative expected value of the second derivative of the log-likelihood with respect to θ , and it measures how sharply peaked the log-likelihood is around its maximum.

#137 · Parametric Estimation

How is the large-sample variance of the MLE approximated using the Fisher information?

Answer

$\text{Var}(\hat{\theta})$ is approximately the reciprocal of the total Fisher information, 1 divided by $I(\theta)$, estimated using either the expected information with $\hat{\theta}$ substituted in or the observed information from the sample data directly.

#138 · Parametric Estimation

What is the general form of an approximate 95 percent confidence interval for an MLE $\hat{\theta}$?

Answer

$\hat{\theta}$ plus or minus 1.96 times the square root of the estimated variance of $\hat{\theta}$, using the standard normal critical value of 1.96 for a 95 percent interval.

#139 · Parametric Estimation

For an exponential MLE with $\hat{\theta} = 10/3$ and $n = 3$, what is the estimated variance of $\hat{\theta}$ using $\text{Var}(\hat{\theta}) = \hat{\theta}^2 / n$?

Answer

$\text{Var}(\hat{\theta})$ equals $(10/3)^2$ divided by 3, which equals 100 divided by 27, approximately 3.7037, giving a standard error of about 1.9245.

#140 · Parametric Estimation

What does the delta method let you do, and what is its variance formula?

Answer

The delta method builds a confidence interval for any smooth function g of the MLE by approximating $\text{Var}(g(\hat{\theta}))$ as $g'(\hat{\theta})^2 \text{Var}(\hat{\theta})$, letting you find the standard error of a transformed quantity like a rate or a percentile.

#141 · Introduction to Credibility

What is the credibility factor Z?

Answer

The weight, between 0 and 1, placed on a risk's own observed experience when blending it with a manual or prior estimate. Z near 1 means the observed data is trusted heavily; Z near 0 means the manual estimate dominates.

#142 · Introduction to Credibility

What is the credibility premium formula?

Answer

Credibility Premium = Z times the observed mean, plus (1 minus Z) times the manual or prior estimate. It is a weighted average of the risk's own data and a broader benchmark.

#143 · Introduction to Credibility

What is the full credibility standard under limited fluctuation credibility?

Answer

$n_{\text{full}} = (z_p / k)^2$, where z_p is the standard normal quantile for confidence level p and k is the tolerance (for example, within 5% of the true mean). It is the number of expected claims needed before data is treated as fully credible.

#144 · Introduction to Credibility

For a 95% confidence level, what standard normal quantile z_p is used in the full credibility formula?

Answer

z_p is approximately 1.96, since 95% confidence leaves 2.5% in each tail, corresponding to the 97.5th percentile of the standard normal distribution.

#145 · Introduction to Credibility

What is the partial credibility formula in limited fluctuation credibility?

Answer

$Z = \sqrt{n / n_{\text{full}}}$, where n is the number of claims actually observed and n_{full} is the full credibility standard. Z is capped at 1.

#146 · Introduction to Credibility

If $n_{\text{full}} = 1,536.64$ and a risk has observed $n = 800$ expected claims, what is Z under limited fluctuation credibility?

Answer

Z equals the square root of $(800 \text{ divided by } 1,536.64)$, which is the square root of 0.52064 , approximately 0.7216 .

#147 · Introduction to Credibility

What does Bühlmann credibility optimize, compared to limited fluctuation credibility?

Answer

Bühlmann credibility chooses Z as the value minimizing expected squared error, a least squares best linear approximation to the risk's true mean, based on the population's actual variance structure, rather than a chosen confidence level and tolerance.

#148 · Introduction to Credibility

What is the Bühlmann credibility factor formula?

Answer

$Z = n / (n + k)$, where n is the number of observed periods (or exposure units) and k is the Bühlmann k parameter, equal to EPV divided by VHM.

#149 · Introduction to Credibility

What does EPV (Expected Value of the Process Variance) measure?

Answer

How much a single risk's own observations vary from period to period purely due to random chance, even if that risk's true underlying mean never changes. It is the average of the within-class variances across all risk classes.

#150 · Introduction to Credibility

What does VHM (Variance of the Hypothetical Means) measure?

Answer

How much the true underlying means genuinely differ from one risk or risk class to another, after removing the part of the spread caused by ordinary sampling noise.

#151 · Introduction to Credibility

What is the Bühlmann k parameter and how does it affect Z ?

Answer

$k = EPV / VHM$. A larger k (more noise relative to genuine differences between risks) lowers Z for any fixed n , since the risk's own data carries less trustworthy signal.

#152 · Introduction to Credibility

How is EPV estimated from a table of r risk classes each observed for n periods?

Answer

EPV_hat is the average, across all r classes, of each class's own sample variance, computed with divisor n minus 1: $EPV_hat = (1/r)$ times the sum over classes of the within-class sample variance.

#153 · Introduction to Credibility

How is VHM estimated, and why does the formula subtract a bias correction term?

Answer

$VHM_hat = [\text{sum of (class mean minus overall mean) squared, divided by (r minus 1)}] \text{ minus (EPV_hat divided by n)}$. The subtraction removes the part of the spread between class means that is caused by ordinary sampling noise rather than genuine differences between risks.

#154 · Introduction to Credibility

Two classes have annual claim counts of 4, 6, 5 (mean 5) and 2, 3, 4 (mean 3), each with within-class sample variance 1. What is EPV_hat?

Answer

EPV_hat is the average of the two within-class variances: (1 plus 1) divided by 2, which equals 1.

#155 · Introduction to Credibility

Using the same two classes (means 5 and 3, overall mean 4, $EPV_{\hat{}} = 1$, $n = 3$), what is $VHM_{\hat{}}$?

Answer

$VHM_{\hat{}} = [(5-4)^2 + (3-4)^2] / (2-1) \text{ minus } (1/3) = 2 \text{ minus } 0.3333$, which equals approximately 1.6667.

#156 · Introduction to Credibility

With $EPV_{\hat{}} = 1$ and $VHM_{\hat{}} = 1.6667$, what is the Bühlmann credibility factor Z for a class with $n = 3$ years of data?

Answer

$k = EPV/VHM = 1/1.6667 = 0.6$, so $Z = n/(n+k) = 3/3.6$, which equals approximately 0.8333.

#157 · Introduction to Credibility

What does Bühlmann-Straub credibility generalize about the plain Bühlmann model?

Answer

It allows exposure (the number of policies, insured lives, or other units underlying each observation) to vary across periods and across risk classes, instead of assuming every risk contributes the same amount of exposure.

#158 · Introduction to Credibility

What is the Bühlmann-Straub credibility factor formula for risk class i ?

Answer

$Z_i = m_i / (m_i + k)$, where m_i is that class's own total exposure (summed across all its periods) and $k = EPV/VHM$, exactly as in the plain Bühlmann model.

#159 · Introduction to Credibility

In Bühlmann-Straub credibility, why can two classes with the same k end up with different credibility factors?

Answer

Because Z_i depends on each class's own total exposure m_i . A class with more total exposure earns a higher Z , even if its underlying k (the ratio of EPV to VHM for the population) is identical to another class's.

#160 · Introduction to Credibility

What is a common mistake to avoid when choosing between the limited fluctuation and Bühlmann partial credibility formulas?

Answer

The limited fluctuation formula uses a square root, $Z = \sqrt{n/n_{\text{full}}}$, while the Bühlmann formula does not, $Z = n/(n+k)$. Mixing the two formulas up is a common error under exam time pressure.

#161 · Pricing and Reserving for Short-Term Coverages

What is the unearned premium reserve (UPR)?

Answer

The portion of collected premium that corresponds to coverage the insurer has not yet provided as of the valuation date. It is a liability because the insurer would owe it back if the policy were cancelled.

#162 · Pricing and Reserving for Short-Term Coverages

What is the pro-rata method for earning premium?

Answer

A method that spreads a policy's total premium evenly, day by day, across the length of the policy term, so premium is earned in direct proportion to the passage of time.

#163 · Pricing and Reserving for Short-Term Coverages

State the UPR formula.

Answer

UPR = Total Premium x (Months Remaining / Total Term in Months), where months remaining is measured from the valuation date to the policy expiration date.

#164 · Pricing and Reserving for Short-Term Coverages

A 12 month policy with premium \$2,400 is effective July 1. What is the UPR at December 31 of the same year?

Answer

6 months have elapsed (July through December) and 6 months remain. $UPR = \$2,400 \times (6/12) = \$1,200$.

#165 · Pricing and Reserving for Short-Term Coverages

What is earned premium, and how does it relate to UPR?

Answer

Earned premium is the portion of premium corresponding to coverage already provided. $\text{Earned Premium} = \text{Total Premium} - \text{UPR}$, since every dollar of premium is either earned or unearned.

#166 · Pricing and Reserving for Short-Term Coverages

What valuation date is used to compute the UPR?

Answer

The date on which the insurer's financial position is being measured. It falls somewhere inside the policy period, and the months remaining are counted from that date to the policy's expiration date.

#167 · Pricing and Reserving for Short-Term Coverages

Why do insurers often assume policies are written mid-month when computing UPR for a large block of business?

Answer

It lets a single simplified formula be applied across an entire block of policies with different effective dates, instead of tracking every policy's exact effective date individually.

#168 · Pricing and Reserving for Short-Term Coverages

Why is the unearned premium reserve considered a liability?

Answer

Because if the policyholder cancelled the policy immediately, the insurer would be obligated to refund the unearned portion of the premium back to the policyholder.

#169 · Pricing and Reserving for Short-Term Coverages

What does IBNR stand for, and what does it represent?

Answer

Incurred But Not Reported. It is the reserve an insurer holds for claims that have already occurred but that the insurer does not yet know about.

#170 · Pricing and Reserving for Short-Term Coverages

What is a case reserve, and how does it differ from IBNR?

Answer

A case reserve is an estimate a claims adjuster sets for a claim that has already been reported, based on that claim's specific facts. IBNR instead covers claims the insurer does not yet know about at all, plus further development on known claims.

#171 · Pricing and Reserving for Short-Term Coverages

What does IBNER stand for?

Answer

Incurred But Not Enough Reported. It refers to additional development expected on claims that are already known and case-reserved, but whose case reserves will likely prove too low or too high.

#172 · Pricing and Reserving for Short-Term Coverages

State the general formula for the unpaid claim estimate.

Answer

Unpaid Claim Estimate = Projected Ultimate Losses minus Paid Losses to Date.

#173 · Pricing and Reserving for Short-Term Coverages

State the formula for IBNR in terms of reported losses.

Answer

IBNR = Projected Ultimate Losses minus Reported Losses, where Reported Losses equals Paid Losses plus Case Reserves.

#174 · Pricing and Reserving for Short-Term Coverages

What does 'reported losses' mean in a reserving context?

Answer

Reported losses equal paid losses plus case reserves. It is the total amount an insurer currently knows about for a group of claims.

#175 · Pricing and Reserving for Short-Term Coverages

Give an example of why a claim might not be reported to an insurer right away.

Answer

A liability claim, such as one arising from a workplace injury, might not surface until a lawsuit is filed years after the event actually occurred.

#176 · Pricing and Reserving for Short-Term Coverages

What role do the chain-ladder, Bornhuetter-Ferguson, and expected claims methods play with respect to IBNR?

Answer

They are three different techniques for estimating the projected ultimate loss figure that IBNR is calculated from, each making different tradeoffs between using the insurer's own reported data and outside expected-loss information.

#177 · Pricing and Reserving for Short-Term Coverages

What is a loss development triangle?

Answer

A table that organizes cumulative claims data by accident year (rows) and development age (columns), shaped like a triangle because older accident years have data at more ages than the most recent accident year.

#178 · Pricing and Reserving for Short-Term Coverages

What is development age?

Answer

The number of months or years that have passed since the start of a given accident year, such as 12 months, 24 months, or 36 months of development.

#179 · Pricing and Reserving for Short-Term Coverages

What is an age-to-age development factor, also called a link ratio?

Answer

A factor measuring how much cumulative losses for a group of accident years grew from one development age to the next, used to project how immature accident years will mature.

#180 · Pricing and Reserving for Short-Term Coverages

How is a volume-weighted average age-to-age factor computed?

Answer

Sum the losses at age $j+1$ across every accident year with data at both age j and age $j+1$, and divide by the sum of the losses at age j for those same accident years.

#181 · Pricing and Reserving for Short-Term Coverages

Two accident years show 12-month losses of 1,000 and 1,100, and 24-month losses of 1,500 and 1,650. What is the 12-to-24 age-to-age factor?

Answer

Factor = $(1,500 + 1,650) / (1,000 + 1,100) = 3,150 / 2,100 = 1.50$.

#182 · Pricing and Reserving for Short-Term Coverages

How does the chain-ladder method project an accident year's ultimate loss?

Answer

Multiply the accident year's latest known cumulative loss by every remaining age-to-age factor out to the oldest age in the triangle.

#183 · Pricing and Reserving for Short-Term Coverages

An accident year's latest cumulative loss is 1,200 at 12 months. The 12-to-24 factor is 1.5 and the 24-to-36 factor is 16/15. What is the chain-ladder ultimate?

Answer

Ultimate = $1,200 \times 1.5 \times (16/15) = 1,800 \times (16/15) = 1,920$.

#184 · Pricing and Reserving for Short-Term Coverages

In a chain-ladder triangle, when is a cumulative loss value treated as the ultimate loss for that accident year?

Answer

When it is observed at the oldest development age in the triangle, meaning losses are assumed to have finished developing by that age.

#185 · Pricing and Reserving for Short-Term Coverages

When does the chain-ladder method work well, and when does it perform poorly?

Answer

It works well when claim reporting and settlement patterns are stable across accident years. It performs poorly when the triangle is thin, since a single large or small claim can distort a link ratio, or after a change in claims handling, business mix, or a large one-time event.

#186 · Pricing and Reserving for Short-Term Coverages

Why can the chain-ladder method be unreliable for a very immature accident year?

Answer

Because the projection multiplies a small reported loss figure by a large development factor, so the resulting ultimate is highly sensitive to random fluctuation in the small amount of data reported so far.

#187 · Pricing and Reserving for Short-Term Coverages

How do you perform a consistency check on a completed chain-ladder projection?

Answer

Sum the individual accident years' projected ultimate losses and separately sum their latest reported losses. The difference between the two totals should equal the sum of the individually computed IBNR figures.

#188 · Pricing and Reserving for Short-Term Coverages

In a triangle, only one accident year has data at both 24 and 36 months, going from 1,500 to 1,600. What is the 24-to-36 age-to-age factor as a fraction?

Answer

$1,600 / 1,500 = 16/15$, or approximately 1.0667.

#189 · Pricing and Reserving for Short-Term Coverages

What is the cumulative development factor (CDF)?

Answer

The product of all remaining age-to-age factors from a given development age out to the oldest age in the triangle. It represents the total remaining growth expected in an accident year's cumulative losses.

#190 · Pricing and Reserving for Short-Term Coverages

What does the Bornhuetter-Ferguson (BF) method do that the chain-ladder method does not?

Answer

It blends the chain-ladder-style projection with an independent, outside a priori estimate of expected losses, rather than relying solely on the reported claims data.

#191 · Pricing and Reserving for Short-Term Coverages

State the BF formula for percentage reported.

Answer

Percentage Reported = $1 / \text{CDF}$, where CDF is the cumulative development factor from the current age out to the oldest age in the triangle.

#192 · Pricing and Reserving for Short-Term Coverages

State the BF ultimate loss formula.

Answer

BF Ultimate = Reported Losses + (Expected Losses) $\times$ (1 minus Percentage Reported).

#193 · Pricing and Reserving for Short-Term Coverages

What is the 'a priori' expected loss estimate used in the BF method?

Answer

An estimate of expected losses made independently of the accident year's own claims experience, often computed as the premium for that year multiplied by an assumed loss ratio from pricing.

#194 · Pricing and Reserving for Short-Term Coverages

If the CDF for an accident year is 1.6, what is the percentage reported?

Answer

Percentage Reported = $1 / 1.6 = 0.625$, or 62.5%.

#195 · Pricing and Reserving for Short-Term Coverages

Reported losses are 1,200, expected losses are 2,000, and percentage reported is 62.5%. What is the BF ultimate?

Answer

$$\text{BF Ultimate} = 1,200 + 2,000 \times (1 - 0.625) = 1,200 + 2,000 \times 0.375 = 1,200 + 750 = 1,950.$$

#196 · Pricing and Reserving for Short-Term Coverages

How is IBNR computed under the BF method?

Answer

BF IBNR = BF Ultimate minus Reported Losses, the same general subtraction used for any ultimate-loss-based reserving method.

#197 · Pricing and Reserving for Short-Term Coverages

As an accident year matures and its percentage reported approaches 100%, what happens to the BF ultimate?

Answer

It converges toward the pure chain-ladder ultimate, since the unreported portion being filled in by the a priori expected losses shrinks toward zero.

#198 · Pricing and Reserving for Short-Term Coverages

Why does the BF method give full weight to actual reported losses?

Answer

Because the reported losses term is added in full and never scaled down; only the unreported portion of ultimate losses is estimated using the outside a priori expected loss figure.

#199 · Pricing and Reserving for Short-Term Coverages

Why is the BF method generally preferred over the pure chain-ladder method for immature accident years?

Answer

Because it reduces the chain-ladder method's sensitivity to random fluctuation in a small amount of reported data, by blending in a stable, independent expected-loss estimate for the unreported portion.

#200 · Pricing and Reserving for Short-Term Coverages

What is the expected claims technique?

Answer

A reserving method that sets the projected ultimate loss equal to the a priori expected losses, entirely ignoring the claims reported so far.

#201 · Pricing and Reserving for Short-Term Coverages

When is the expected claims technique most useful?

Answer

When there is little or no reliable claims experience available, such as for a brand-new line of business or an accident year with almost no reported claims yet.

#202 · Pricing and Reserving for Short-Term Coverages

How is IBNR computed under the expected claims technique?

Answer

IBNR equals the full a priori expected loss figure minus whatever has been reported so far, since the projected ultimate is simply set equal to the expected losses.

#203 · Pricing and Reserving for Short-Term Coverages

What is the main tradeoff of the expected claims technique?

Answer

It is completely insulated from random noise in a thin or immature triangle, but it also completely ignores any genuine signal contained in claims that have actually been reported.

#204 · Pricing and Reserving for Short-Term Coverages

How does the BF method relate to the expected claims technique and the chain-ladder method as limiting cases?

Answer

When percentage reported is 0%, the BF formula reduces to the expected claims technique. When percentage reported is 100%, it reduces to the chain-ladder method.

#205 · Pricing and Reserving for Short-Term Coverages

As an accident year ages, which reserving method does an actuary typically move through in sequence?

Answer

From the expected claims technique, to the Bornhuetter-Ferguson method, to a pure chain-ladder projection, as the accident year's own reported data becomes more credible.

#206 · Pricing and Reserving for Short-Term Coverages

What is the pure premium in ratemaking, and what is another name for it?

Answer

The average loss per exposure unit, also called the loss cost. $\text{Pure Premium} = \text{Total Projected Ultimate Losses} / \text{Total Exposures}$.

#207 · Pricing and Reserving for Short-Term Coverages

Does the pure premium cover an insurer's expenses or profit?

Answer

No. The pure premium covers expected losses only. Expenses and profit are added separately through expense loading.

#208 · Pricing and Reserving for Short-Term Coverages

Total ultimate losses are \$1,200,000 across 2,000 exposures. What is the pure premium?

Answer

Pure Premium = $\$1,200,000 / 2,000 = \600 per exposure.

#209 · Pricing and Reserving for Short-Term Coverages

What is the loss ratio method used for?

Answer

Adjusting an existing, already-charged rate up or down, based on how the actual experienced loss ratio compares to the permissible loss ratio built into that rate.

#210 · Pricing and Reserving for Short-Term Coverages

What is the permissible loss ratio?

Answer

The portion of premium an insurer planned to have available for losses, after covering its expense and profit provisions, when the current rate was originally built.

#211 · Pricing and Reserving for Short-Term Coverages

State the rate change factor formula used in the loss ratio method.

Answer

Rate Change Factor = Experienced Loss Ratio / Permissible Loss Ratio.

#212 · Pricing and Reserving for Short-Term Coverages

If the rate change factor is greater than 1.0, what does that indicate?

Answer

The current rate is too low relative to the losses actually being generated, so a rate increase is indicated.

#213 · Pricing and Reserving for Short-Term Coverages

An insurer's experienced loss ratio is 70% and its permissible loss ratio is 65%. What is the indicated rate change factor?

Answer

Rate Change Factor = $0.70 / 0.65 = 1.0769$, indicating about a 7.69% rate increase.

#214 · Pricing and Reserving for Short-Term Coverages

A current rate of \$500 needs a rate change factor of 1.0769 applied. What is the new rate?

Answer

New Rate = $\$500 \times 1.0769 = \538.46 .

#215 · Pricing and Reserving for Short-Term Coverages

What are fixed expenses in ratemaking, and how are they expressed?

Answer

Costs that do not vary with the size of the policy, such as a flat per-policy underwriting cost, expressed as a dollar amount per exposure.

#216 · Pricing and Reserving for Short-Term Coverages

What are variable expenses in ratemaking, and how are they expressed?

Answer

Costs that scale with premium, such as commissions and premium taxes, expressed as a percentage of the final rate rather than as a dollar amount.

#217 · Pricing and Reserving for Short-Term Coverages

How is the profit provision typically expressed in the rate formula?

Answer

As a percentage of the final rate, added on top of the loss and expense provisions so the insurer earns a reasonable return.

#218 · Pricing and Reserving for Short-Term Coverages

State the expense-loaded rate formula.

Answer

Rate = (Pure Premium + Fixed Expense per Exposure) / (1 - Variable Expense % - Profit %).

#219 · Pricing and Reserving for Short-Term Coverages

Why must variable expenses and profit be divided out of the rate rather than simply added on top?

Answer

Because they are stated as a percentage of the final rate itself, not as a percentage of the pure premium, so the rate must be solved for algebraically to leave enough behind after those percentages are removed.

#220 · Pricing and Reserving for Short-Term Coverages

Pure premium plus fixed expense per exposure is \$650, variable expenses are 20% of rate, and profit is 5% of rate. What is the rate?

Answer

Rate = $\$650 / (1 - 0.20 - 0.05) = \$650 / 0.75 = \$866.67$.

#221 · Option Pricing Fundamentals

What is the payoff of a European call option at expiration?

Answer

$\max(S_T - K, 0)$, where S_T is the stock price at expiration and K is the strike price. The owner only exercises when S_T exceeds K .

#222 · Option Pricing Fundamentals

What is the payoff of a European put option at expiration?

Answer

$\max(K - S_T, 0)$, where K is the strike price and S_T is the stock price at expiration. The owner only exercises when S_T is below K .

#223 · Option Pricing Fundamentals

State the put-call parity equation.

Answer

$C - P = S_0 - K \cdot v^T$, where C is the call price, P is the put price, S_0 is the current stock price, and $K \cdot v^T$ is the present value of the strike.

#224 · Option Pricing Fundamentals

Why must put-call parity hold regardless of which pricing model is used?

Answer

Because a portfolio of a call plus a bond paying K at time T has the identical payoff, in every future outcome, as a portfolio of a put plus one share of stock. Two portfolios with identical future payoffs must have identical prices today, or a riskless arbitrage profit would exist.

#225 · Option Pricing Fundamentals

What discounting rule must always be followed when applying put-call parity?

Answer

Use one discounting convention consistently: a continuously compounded rate r requires $K \cdot e^{-rT}$, while an annual effective rate i requires $K \cdot (1+i)^{-T}$. Mixing the two conventions in one calculation is a common source of errors.

#226 · Option Pricing Fundamentals

In the one-period binomial model, what are u and d ?

Answer

u is the factor by which the stock price multiplies in the up state, and d is the factor by which it multiplies in the down state, with $d < 1 < u$.

#227 · Option Pricing Fundamentals

What is the formula for the risk-neutral probability p^* under continuous compounding?

Answer

$p^* = (e^{r^*h} - d) / (u - d)$, where r is the continuously compounded risk-free rate and h is the length of one period.

#228 · Option Pricing Fundamentals

What is the formula for the risk-neutral probability p^* under discrete compounding?

Answer

$p^* = (1 + r - d) / (u - d)$, where r is the per-period effective risk-free rate.

#229 · Option Pricing Fundamentals

Once p^* is known, how is a one-period option price calculated?

Answer

$C_0 = v * [p^* * C_u + (1 - p^*) * C_d]$, where v is the one-period discount factor and C_u , C_d are the option payoffs in the up and down states.

#230 · Option Pricing Fundamentals

Does the real-world probability of the stock going up ever enter the binomial pricing formula?

Answer

No. Only u , d , the risk-free rate, and the strike price are needed. The risk-neutral probability p^* replaces the real-world probability entirely.

#231 · Option Pricing Fundamentals

A stock at 50 moves to 60 (u factor 1.20) or 45 (d factor 0.90) over one year. The continuously compounded rate is 5%. Find the risk-neutral probability p^* .

Answer

$$p^* = (e^{0.05} - 0.90) / (1.20 - 0.90) = (1.051271 - 0.90) / 0.30 = 0.504237.$$

#232 · Option Pricing Fundamentals

Using the prior card's setup, find the price of a call with strike 50.

Answer

$$C_u = \max(60-50, 0) = 10, C_d = \max(45-50, 0) = 0. \\ C_0 = e^{-0.05} * [0.504237*10 + 0.495763*0] = 0.951229 * 5.04237 = 4.80.$$

#233 · Option Pricing Fundamentals

What is backward induction in a multi-period binomial tree?

Answer

Working from the final expiration nodes back to today, one period at a time, applying the same one-period risk-neutral pricing formula at each node until a single price emerges at the starting node.

#234 · Option Pricing Fundamentals

In an n -period binomial tree, how many distinct final stock prices are possible?

Answer

$n+1$, because the ending price depends only on the number of up-moves out of n total moves, not the order in which they occurred.

#235 · Option Pricing Fundamentals

A two-period tree has $S_0 = 100$, $u = 1.10$, $d = 0.95$, $r = 6\%$ per period. Find p^* .

Answer

$$p^* = (1 + 0.06 - 0.95) / (1.10 - 0.95) = 0.11 / 0.15 = 0.7333.$$

#236 · Option Pricing Fundamentals

In the prior two-period tree, the call payoffs at expiration are $C_{uu} = 21$, $C_{ud} = 4.50$, $C_{dd} = 0$. Find the call price one period earlier at the up node.

Answer

$$C_u = [p^*C_{uu} + (1-p^*)C_{ud}] / (1+r) = [0.7333*21 + 0.2667*4.50] / 1.06 = 16.60 / 1.06 = 15.66.$$

#237 · Option Pricing Fundamentals

What are the Black-Scholes formulas for a European call and put?

Answer

Call: $C = S_0 N(d_1) - K e^{-rT} N(d_2)$. Put: $P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$, using the same d_1 and d_2 in both.

#238 · Option Pricing Fundamentals

What are the formulas for d_1 and d_2 in Black-Scholes?

Answer

$d_1 = [\ln(S_0/K) + (r + \sigma^2/2)T] / (\sigma \sqrt{T})$, and $d_2 = d_1 - \sigma \sqrt{T}$.

#239 · Option Pricing Fundamentals

What does $N(x)$ represent in the Black-Scholes formula?

Answer

The cumulative standard normal distribution function: the probability that a standard normal random variable is less than or equal to x , read from a standard normal table.

#240 · Option Pricing Fundamentals

What is the economic interpretation of $N(d_2)$ in the Black-Scholes call formula?

Answer

$N(d_2)$ is the risk-neutral probability that the call finishes in the money, meaning the stock price at expiration ends up above the strike.

#241 · Option Pricing Fundamentals

For $S_0 = K = 50$, $r = 5\%$, $\sigma = 30\%$, $T = 1$, find $d1$ and $d2$.

Answer

$$d1 = [0 + (0.05 + 0.045) \cdot 1] / 0.30 = 0.095 / 0.30 = 0.3167. \quad d2 = 0.3167 - 0.30 = 0.0167.$$

#242 · Option Pricing Fundamentals

Using $d1 = 0.3167$ and $d2 = 0.0167$ with $N(d1) \approx 0.6242$ and $N(d2) \approx 0.5067$, find the Black-Scholes call price for $S_0 = K = 50$, $r = 5\%$, $T = 1$.

Answer

$$C = 50 \cdot 0.6242 - 50 \cdot e^{-0.05} \cdot 0.5067 = 31.21 - 47.56 \cdot 0.5067 = 31.21 - 24.10 = 7.11.$$

#243 · Option Pricing Fundamentals

What single idea unifies the binomial model and the Black-Scholes formula?

Answer

Risk-neutral valuation: the price of any derivative equals its expected payoff under a risk-neutral probability measure, discounted at the risk-free rate. The binomial tree converges to the Black-Scholes price as the number of periods grows large.

#244 · Option Pricing Fundamentals

How does a guaranteed minimum maturity benefit (GMMB) behave like an option?

Answer

The policyholder receives $\max(A_T, G) = A_T + \max(G - A_T, 0)$, which is the account value plus the payoff of a European put option on the fund with strike G . The insurer has effectively sold a put option bundled with the fund.

#245 · Option Pricing Fundamentals

How does a guaranteed minimum death benefit differ from a GMMB in option terms?

Answer

Both behave like a put option on the underlying fund with the guaranteed amount as the strike, but the death benefit's put is exercised at an uncertain, mortality-driven date rather than a fixed maturity date.

#246 · Long-Term Coverages and Retirement Programs

What is whole life insurance?

Answer

A policy that pays a death benefit whenever the insured dies, with no expiration date, so the only uncertainty is the timing of the payment, not whether it occurs.

#247 · Long-Term Coverages and Retirement Programs

What is term life insurance?

Answer

A policy that pays a death benefit only if the insured dies within a specified term, such as 10, 20, or 30 years. If the insured survives past the term, no benefit is paid.

#248 · Long-Term Coverages and Retirement Programs

What is endowment insurance?

Answer

A policy that pays a benefit if the insured dies within the term, and also pays a maturity benefit if the insured survives to the end of the term, so a payment is guaranteed either way.

#249 · Long-Term Coverages and Retirement Programs

What is a pure endowment?

Answer

A coverage that pays a benefit only if the insured survives to the end of a specified period, and pays nothing if the insured dies before then. It is the mirror image of term insurance.

#250 · Long-Term Coverages and Retirement Programs

How is endowment insurance related to term insurance and a pure endowment?

Answer

Endowment insurance is mathematically the combination of a term insurance policy plus a pure endowment written over the same period; their payment streams add together.

#251 · Long-Term Coverages and Retirement Programs

What is deferred insurance?

Answer

A coverage where the benefit trigger cannot occur until after a waiting period, called the deferral period, has passed. No benefit is paid for a death during the deferral period.

#252 · Long-Term Coverages and Retirement Programs

What is the key difference between a term policy and a deferred policy?

Answer

A term policy sets an ending point after which coverage stops, while a deferred policy sets a starting point before which coverage has not yet begun.

#253 · Long-Term Coverages and Retirement Programs

What is an immediate annuity?

Answer

An annuity where payments begin right away, typically with the first payment due immediately or within one payment period of purchase.

#254 · Long-Term Coverages and Retirement Programs

What is a deferred annuity?

Answer

An annuity where the start of payments is delayed until a future date, often chosen to line up with retirement age, with no payments made during the deferral period.

#255 · Long-Term Coverages and Retirement Programs

What is the difference between a whole life annuity and a temporary annuity?

Answer

A whole life annuity continues paying for as long as the annuitant is alive, with no fixed end date. A temporary annuity pays only for a fixed maximum number of years, or until earlier death.

#256 · Long-Term Coverages and Retirement Programs

What is a defined benefit (DB) plan?

Answer

A retirement plan that promises a specific benefit formula agreed upon in advance, usually a percentage of salary times years of service, with the employer bearing the investment risk.

#257 · Long-Term Coverages and Retirement Programs

What is a defined contribution (DC) plan?

Answer

A retirement plan that defines how much money is contributed to an employee's account each period, with the eventual benefit depending on investment performance. The employee bears the investment risk.

#258 · Long-Term Coverages and Retirement Programs

What is the accrued benefit in a DB plan?

Answer

The portion of the eventual pension an employee has earned based on service completed so far, calculated by applying the plan's benefit formula to service and salary history to date.

#259 · Long-Term Coverages and Retirement Programs

What does vesting mean in a DB plan?

Answer

Vesting is the point at which an employee gains a guaranteed, non-forfeitable right to their accrued benefit, even if they leave the employer before retirement age.

#260 · Long-Term Coverages and Retirement Programs

What is normal retirement age (NRA)?

Answer

The age specified in a DB plan at which an employee is entitled to receive their full accrued benefit with no reduction.

#261 · Long-Term Coverages and Retirement Programs

Why does a DB plan apply an early retirement reduction factor?

Answer

Because retiring before normal retirement age means payments start earlier and are expected to continue over more years, so the plan reduces the benefit amount to compensate for the longer expected payment period.

#262 · Long-Term Coverages and Retirement Programs

How is the replacement ratio calculated?

Answer

Replacement ratio equals retirement income divided by pre-retirement salary. For example, a \$31,500 pension against an \$88,000 final salary gives a replacement ratio of about 35.8 percent.

#263 · Long-Term Coverages and Retirement Programs

How do you compute a DB accrued benefit using a percent-times-final-average-salary-times-service formula?

Answer

Multiply the benefit percentage by final average salary and by years of service. Example: 1.5 percent times \$84,000 final average salary times 25 years of service equals \$31,500 per year.

#264 · Long-Term Coverages and Retirement Programs

What is the actuarial liability of a DB plan?

Answer

The actuarial present value of all promised future benefit payments to current and former employees, requiring assumptions about mortality, salary growth, interest rates, and turnover.

#265 · Long-Term Coverages and Retirement Programs

What is the liability of a DC plan?

Answer

Simply the total of the account balances currently held for all participants, since the plan never promised a specific future benefit, only contributions plus investment earnings.

#266 · Mortality Models

What does T_x represent?

Answer

T_x is the future lifetime random variable for a life aged exactly x . It is the number of complete and partial years this person will live beyond age x .

#267 · Mortality Models

If a life aged 40 dies at exact age 75.3, what is T_{40} ?

Answer

$T_{40} = 35.3$, since the person lived 35.3 more years past age 40.

#268 · Mortality Models

What is the difference between T_x and T_0 ?

Answer

T_0 is the future lifetime of a newborn (age at death). T_x is the future lifetime of a life who has already reached age x , conditional on surviving to age x .

#269 · Mortality Models

Why does most mortality theory start with T_0 and then adapt to T_x ?

Answer

Because insurers issue policies to lives who are already alive at some age x , so the newborn-based theory must be conditioned on survival to x before it applies to a real policyholder.

#270 · Mortality Models

Define the survival function $S(x)$.

Answer

$S(x)$ is the probability that a newborn survives to at least age x , so $S(x) = P(T_0 > x)$.

#271 · Mortality Models

What are the boundary values of $S(x)$?

Answer

$S(0) = 1$, because everyone is alive at birth, and $S(x)$ approaches 0 as x approaches infinity, since no one lives forever.

#272 · Mortality Models

Is $S(x)$ increasing or decreasing, and why?

Answer

$S(x)$ is decreasing, because survival to a later age can never be more likely than survival to an earlier age.

#273 · Mortality Models

How is the probability of dying before age x related to $S(x)$?

Answer

The probability that a newborn dies before age x is 1 minus $S(x)$.

#274 · Mortality Models

Write the formula for ${}_nPx$ using the survival function.

Answer

${}_nPx = S(x+n) / S(x)$. This divides the unconditional survival probability by $S(x)$ to condition on already being alive at age x .

#275 · Mortality Models

Why must nPx be computed by dividing by $S(x)$ rather than using $S(x+n)$ alone?

Answer

$S(x+n)$ alone is an unconditional probability measured from birth. Dividing by $S(x)$ conditions on the life already being alive at age x , which is what insurers actually need.

#276 · Mortality Models

Define the force of mortality μ_x .

Answer

$\mu_x = -S'(x) / S(x)$. It is the instantaneous rate of dying at exact age x , given survival to that age.

#277 · Mortality Models

Why is μ_x always positive even though $S'(x)$ is negative?

Answer

$S(x)$ is decreasing so $S'(x)$ is negative. The formula for μ_x has a minus sign in front, which turns that negative derivative into a positive value.

#278 · Mortality Models

What field outside actuarial science uses a concept equivalent to μ_x ?

Answer

General survival analysis, where μ_x is called the hazard rate.

#279 · Mortality Models

Write $S(x)$ in terms of the integral of the force of mortality.

Answer

$S(x) = \exp$ of negative the integral from 0 to x of $\mu_t dt$.

#280 · Mortality Models

Why is the integral formula for $S(x)$ important for parametric mortality laws?

Answer

Many mortality laws, such as Gompertz and Makeham, specify μ_x directly as a formula, and this integral is used to derive $S(x)$ and every survival probability from that formula.

#281 · Mortality Models

Under a constant force of mortality μ over the interval x to $x+n$, what is ${}_nPx$?

Answer

${}_nPx = \exp(\text{negative } \mu \text{ times } n).$

#282 · Mortality Models

When is the constant force of mortality assumption typically used?

Answer

As a simplifying device inside larger calculations, particularly when the interval involved is a fraction of a year and no other information is given.

#283 · Mortality Models

What is a life table radix?

Answer

The radix is the arbitrary starting number of lives assumed at age 0 in a life table, commonly a round number such as 100,000.

#284 · Mortality Models

Define I_x in terms of the radix and $S(x)$.

Answer

$I_x = I_0 \text{ times } S(x)$, the expected number of survivors to age x out of the I_0 lives who started at birth.

#285 · Mortality Models

Is l_x a probability?

Answer

No. l_x is an expected count of lives, not a probability, although it is built directly from the survival probability $S(x)$.

#286 · Mortality Models

Define d_x .

Answer

d_x is the expected number of deaths between age x and age $x+1$, equal to l_x minus l_{x+1} .

#287 · Mortality Models

Define q_x and give its formula in terms of d_x and l_x .

Answer

q_x is the probability that a life aged exactly x dies within one year. $q_x = d_x / l_x$.

#288 · Mortality Models

Define p_x and give two equivalent formulas for it.

Answer

p_x is the probability a life aged exactly x survives at least one more year. $p_x = 1 - q_x = l_{x+1} / l_x$.

#289 · Mortality Models

Given $l_{60} = 100,000$ and $l_{61} = 99,000$, find q_{60} .

Answer

$$d_{60} = 100,000 - 99,000 = 1,000. \quad q_{60} = 1,000 / 100,000 = 0.01000.$$

#290 · Mortality Models

Given $q_{60} = 0.01000$, find p_{60} .

Answer

$$p_{60} = 1 - 0.01000 = 0.99000.$$

#291 · Mortality Models

Why are q_x and p_x described as the building blocks of insurance pricing?

Answer

A one year term insurance on a life aged x pays out with probability q_x and pays nothing with probability p_x , and every more complex insurance and annuity formula is built by chaining q_x and p_x values across many years.

#292 · Mortality Models

Write nPx using life table notation.

Answer

$$nPx = l_{x+n} / l_x.$$

#293 · Mortality Models

Write nPx as a chain of one year survival probabilities.

Answer

$nPx = p_x \text{ times } p_{x+1} \text{ times } p_{x+2} \text{ times } \dots \text{ times } p_{x+n-1}.$

#294 · Mortality Models

Given $l_{60} = 100,000$ and $l_{63} = 96,500$, find $3P60$.

Answer

$3P60 = l_{63} / l_{60} = 96,500 / 100,000 = 0.96500.$

#295 · Mortality Models

Define the n-year deferred probability of death ${}_n|q_x$.

Answer

${}_n|q_x$ is the probability a life aged x survives n years and then dies in the following year. ${}_n|q_x = {}_nPx$ times $q_{x+n} = (l_{x+n} - l_{x+n+1}) / l_x$.

#296 · Mortality Models

Define ${}_n|m q_x$.

Answer

${}_n|m q_x$ is the probability a life aged x survives n years and then dies within the following m years. ${}_n|m q_x = {}_nPx - {}_{n+m}Px = (l_{x+n} - l_{x+n+m}) / l_x$.

#297 · Mortality Models

What is a plain English way to remember the $n|mqx$ formula?

Answer

It is the number alive at the start of the deferral period minus the number still alive at the end of the whole period, divided by the number alive at age x .

#298 · Mortality Models

Given $l_{60} = 100,000$, $l_{61} = 99,000$, and $l_{62} = 97,800$, find q_{61} .

Answer

$d_{61} = 99,000 - 97,800 = 1,200$. $q_{61} = 1,200 / 99,000 = 0.012121$.

#299 · Mortality Models

Define the curtate future lifetime K_x .

Answer

K_x is the integer number of complete future years lived by a life aged x . $K_x = \text{floor}(T_x)$, the greatest whole number less than or equal to T_x .

#300 · Mortality Models

If $T_x = 35.3$, what is K_x ?

Answer

$K_x = 35$, since $\text{floor}(35.3) = 35$.

#301 · Mortality Models

Give the probability formula $P(K_x = k)$.

Answer

$P(K_x = k) = {}_k p_x$ times q_{x+k} , the probability of surviving k full years and then dying in the following year.

#302 · Mortality Models

Define the complete expectation of life $e\text{-degree-}x$.

Answer

$e\text{-degree-}x$ is the expected value of T_x , the average number of years (counting partial years) a life aged x is expected to live. $e\text{-degree-}x =$ the integral from 0 to infinity of $t p_x dt$.

#303 · Mortality Models

Define the curtate expectation of life e_x .

Answer

e_x is the expected value of K_x , the average number of complete future years lived. $e_x =$ the sum from $k=1$ to infinity of kP_x .

#304 · Mortality Models

In plain English, how is e_x computed from a life table?

Answer

Add together the probability of surviving 1 more year, plus the probability of surviving 2 more years, plus the probability of surviving 3 more years, and so on, for every future year.

#305 · Mortality Models

Why does the sum for e_x have only finitely many nonzero terms in a table with a maximum age ω ?

Answer

Because $kPx = 0$ once $x+k$ exceeds ω , since l_x is zero beyond the table's terminal age.

#306 · Mortality Models

A life table has $l_{60} = 100,000$, $l_{61} = 99,000$, $l_{62} = 97,800$, $l_{63} = 96,500$, $l_{64} = 95,100$, $l_{65} = 0$. Find $1P60$ through $5P60$.

Answer

$1P60 = 0.99000$, $2P60 = 0.97800$, $3P60 = 0.96500$,
 $4P60 = 0.95100$, $5P60 = 0.00000$.

#307 · Mortality Models

Using the table where $1P60 = 0.99000$, $2P60 = 0.97800$, $3P60 = 0.96500$, $4P60 = 0.95100$, $5P60 = 0.00000$, find e_{60} .

Answer

$$e_{60} = 0.99000 + 0.97800 + 0.96500 + 0.95100 + 0.00000 = 3.884 \text{ years.}$$

#308 · Mortality Models

How does e_x differ from $e\text{-degree-}x$ in what it represents?

Answer

e_x expects only complete future years lived (a sum over discrete kP_x values), while $e\text{-degree-}x$ expects the exact continuous future lifetime T_x , including partial years (an integral over continuous tP_x values).

#309 · Mortality Models

Which expectation of life, e_x or $e_{\text{degree-}x}$, is most often used directly in FAM life table calculations?

Answer

e_x , the curtate expectation, since it can be computed from a life table using only addition once each kP_x is known.

#310 · Mortality Models

What is select mortality?

Answer

Mortality that depends on both attained age and how recently the life was medically underwritten, since a recently examined life tends to be healthier than average for that age.

#311 · Mortality Models

Why does select mortality fade over time into ultimate mortality?

Answer

The extra health information gained from underwriting becomes stale as years pass, so mortality gradually converges back to the general population level for that age.

#312 · Mortality Models

What does the notation $q_{[x]}$ mean?

Answer

It is the probability of death in the first year after selection (underwriting) at age x .

#313 · Mortality Models

What does the notation $q_{\{x\}+1}$ mean?

Answer

It is the probability of death in the second year after selection at age x , when the life is now one year older.

#314 · Mortality Models

What is the select period?

Answer

The fixed number of years after underwriting during which selection is assumed to still have an effect on mortality.

#315 • Mortality Models

If the select period has length r , what equation links select and ultimate rates?

Answer

$q_{[x]+r} = q_{x+r}$. Select and ultimate rates agree exactly once r years have passed since underwriting.

#316 • Mortality Models

How are select and ultimate life tables typically laid out?

Answer

With one column per duration since selection (0, 1, 2, and so on through the select period) followed by a final column of ultimate rates indexed by attained age.

#317 · Mortality Models

What is a common careless error when reading a select and ultimate table?

Answer

Reading straight down an attained-age column out of habit instead of choosing the select column that matches how long ago the life was underwritten.

#318 · Mortality Models

What does SULT stand for?

Answer

The Standard Ultimate Life Table, the reference mortality table supplied to candidates on the FAM exam.

#319 · Mortality Models

What is the radix of the SULT?

Answer

$l_0 = 100,000$ lives.

#320 · Mortality Models

Does the SULT have a select period?

Answer

No. The SULT is an ultimate table, meaning every mortality rate in it depends on attained age alone.

#321 · Mortality Models

What skill does the FAM exam test with SULT-based questions?

Answer

Fluent use of the table itself: looking up l_x at the correct ages, combining l_x entries into multi-year and deferred probabilities, and summing survival probabilities to get life expectancies.

#322 · Mortality Models

What extra feature does the SULT incorporate beyond a basic age-based mortality pattern?

Answer

An assumption about mortality improvement over calendar time, since later generations tend to experience lower mortality at a given age than earlier generations did.

#323 · Mortality Models

Why is careful reading of ages in the SULT especially important?

Answer

Because the SULT is a fixed published table rather than a formula, most SULT-based questions reduce to careful lookups and arithmetic, so misreading which age an l_x entry corresponds to is a common source of error.

#324 · Mortality Models

State Gompertz's law for the force of mortality.

Answer

$\mu_x = B \text{ times } c \text{ to the power } x$, where B and c are positive constants and c is greater than 1.

#325 · Mortality Models

In Gompertz's law, what do the parameters B and c control?

Answer

B sets the overall scale of mortality, and c controls how quickly mortality rises with age; a larger c means mortality increases faster as a person gets older.

#326 · Mortality Models

State Makeham's law for the force of mortality.

Answer

$\mu_x = A + B \text{ times } c \text{ to the power } x$, extending Gompertz's law with an added age-independent constant A.

#327 · Mortality Models

What does the constant A represent in Makeham's law?

Answer

An age-independent component of mortality, such as background accident risk, that does not grow with age the way biological aging risk does.

#328 · Mortality Models

What happens to Makeham's law when $A = 0$?

Answer

It reduces exactly to Gompertz's law, so Gompertz's law is a special case of Makeham's law.

#329 · Mortality Models

Under Gompertz's law with $B = 0.0001$ and $c = 1.1$, find μ_{70} .

Answer

$\mu_{70} = 0.0001$ times 1.1 to the power 70 . Since 1.1 to the power 70 is approximately 789.75 , μ_{70} is approximately 0.0790 .

#330 · Mortality Models

What is the closed form survival function under Gompertz's law?

Answer

$S(x) = \exp$ of negative (B divided by the natural log of c) times (c to the power x minus 1).

#331 · Present Value Random Variables

What is K_x ?

Answer

K_x is the curtate future lifetime of a person age x , the number of complete years the person lives beyond age x before dying. If death happens during the $(k+1)$ th year, K_x equals k .

#332 · Present Value Random Variables

Define the present value random variable Z for a discrete whole life insurance of 1.

Answer

$Z = v^{(K_x+1)}$, where $v = 1/(1+i)$ is the annual discount factor. The benefit of 1 is paid at the end of the year of death, discounted back that many years.

#333 · Present Value Random Variables

Why is the exponent on Z written as K_{x+1} and not K_x ?

Answer

Because the benefit is paid at the end of the year of death, not the start. If $K_x = 0$ (death in year one), the payment happens at time 1, so the discount factor is v^1 .

#334 · Present Value Random Variables

What is A_x ?

Answer

A_x is the actuarial present value of a whole life insurance of 1, defined as $A_x = E[Z]$. It is the expected present value of the death benefit.

#335 · Present Value Random Variables

Write the summation formula for A_x .

Answer

$A_x = \sum_{k=0}^{\infty} v^{k+1} \cdot {}_k p_x \cdot q_{x+k}$, where ${}_k p_x$ is the probability of surviving k years and q_{x+k} is the probability of dying in the following year.

#336 · Present Value Random Variables

What does ${}_k p_x$ mean?

Answer

${}_k p_x$ is the probability that a person currently age x survives at least k more years, that is, survives to age $x+k$.

#337 · Present Value Random Variables

What does q_{x+k} mean?

Answer

q_{x+k} is the probability that a person who has already reached age $x+k$ dies within the following one year.

#338 · Present Value Random Variables

What does the product $k p_x$ times q_{x+k} represent?

Answer

It is the probability that a person age x survives k full years and then dies during year $k+1$, which is exactly the probability that K_x equals k .

#339 · Present Value Random Variables

What is term insurance, and how is its actuarial present value notated?

Answer

Term insurance pays a benefit only if death occurs within a fixed n -year period. It is written $A^1_{x:n}$, with the superscript 1 marking that the benefit is contingent on death.

#340 · Present Value Random Variables

Write the formula for $A^1_{x:n}$, an n -year term insurance.

Answer

$A^1_{x:n} = \sum_{k=0}^{n-1} v^{k+1} \cdot {}_k p_x \cdot q_{x+k}$. It is the whole life sum truncated after n years.

#341 • Present Value Random Variables

What is a pure endowment, and how is its actuarial present value notated?

Answer

A pure endowment pays 1 only if the insured survives to the end of an n -year period. Its notation places the superscript 1 below and to the right of the A , written $A^{\text{1}}_{x:n}$, sometimes also written nEx .

#342 • Present Value Random Variables

Write the formula for the pure endowment $A^{\text{1}}_{x:n}$.

Answer

$A^{\text{1}}_{x:n} = v^n \text{ times } npx$, the discounted value of 1 multiplied by the probability of surviving all n years.

#343 · Present Value Random Variables

What is endowment insurance, and how is its actuarial present value found?

Answer

Endowment insurance pays 1 whether death occurs within n years or the person survives to the end. Its actuarial present value is $A_{x:n} = A^1_{x:n} + A^{\{1\}}_{x:n}$, the sum of the term insurance and the pure endowment.

#344 · Present Value Random Variables

Does an endowment insurance always pay a benefit?

Answer

Yes. Either the death benefit is paid if death occurs during the term, or the survival benefit is paid if the person is alive at the end of the term. One or the other always happens.

#345 · Present Value Random Variables

What is a deferred insurance?

Answer

A deferred insurance pays a death benefit only if death occurs after a waiting period of m years. Its formula starts the summation at $k = m$ using the same v^{k+1} times $k p_x$ times q_{x+k} terms as whole life insurance, with no extra factor needed since $k p_x$ already reflects survival to that year; equivalently it can be written as v^m times $m p_x$ times A_{x+m} .

#346 · Present Value Random Variables

If a term insurance pays nothing, what must have happened?

Answer

The insured survived past the end of the n -year term. Term insurance only pays a benefit if death occurs strictly within the n -year period.

#347 · Present Value Random Variables

If a pure endowment pays nothing, what must have happened?

Answer

The insured died at some point before the end of the n -year period. A pure endowment pays only on survival to exactly time n .

#348 · Present Value Random Variables

Define the present value random variable Y for a whole life annuity-due.

Answer

$Y = a\text{-}\overline{\angle}(K_x+1)$, the present value of an annuity-certain paying 1 at the start of each year for K_x+1 years, the number of payments actually received before death.

#349 · Present Value Random Variables

Why does an annuitant who dies with $K_x = k$ receive $k+1$ payments, not k ?

Answer

Payments happen at times 0, 1, 2, up through k (annuity-due, paid at the start of each year), and the person is alive at time 0 by assumption, so they collect $k+1$ total payments before dying.

#350 · Present Value Random Variables

Write the closed-form formula for the annuity-certain $a_{\overline{n}|i}$.

Answer

$a_{\overline{n}|i} = (1 - v^n) / d$, where d is the annual discount rate.

#351 · Present Value Random Variables

How is the discount rate d related to i and v ?

Answer

$d = 1 - v = i/(1+i) = i \text{ times } v$. It represents the amount deducted from 1 unit paid at year end to find its value today.

#352 · Present Value Random Variables

What is $\ddot{a}_x$?

Answer

$\ddot{a}_x$ is the actuarial present value of a whole life annuity-due, defined as $\ddot{a}_x = E[Y]$. The two dots signal payments at the start of each year rather than the end.

#353 · Present Value Random Variables

What is $\ddot{a}_{x:n}$?

Answer

$\ddot{a}_{x:n}$ is the actuarial present value of a temporary annuity-due, paying 1 at the start of each year for at most n years, stopping at death or at n years, whichever comes first.

#354 · Present Value Random Variables

Express Y directly in terms of Z .

Answer

$Y = (1 - Z) / d$, since both Y and Z are built from the same random variable K_x through $v^{(K_x+1)}$.

#355 • Present Value Random Variables

State the key relationship between A_x and $\ddot{a}_x$.

Answer

$A_x = 1 - d \text{ times } \ddot{a}_x$, which follows directly from taking the expected value of $Y = (1-Z)/d$.

#356 • Present Value Random Variables

Rearrange $A_x = 1 - d \text{ times } \ddot{a}_x$ to solve for $\ddot{a}_x$.

Answer

$\ddot{a}_x = (1 - A_x) / d$.

#357 · Present Value Random Variables

Does the identity $A_x = 1 - d \text{ times } \ddot{a}_x$ also hold for temporary and endowment insurance?

Answer

Yes, provided the insurance and annuity subscripts match exactly, for example $A_{x:n} = 1 - d \text{ times } \ddot{a}_{x:n}$, since both must be built from the same underlying random variable over the same period.

#358 · Present Value Random Variables

Why can't you mix an n-year term insurance with a whole life annuity in the $A = 1 - d \cdot \ddot{a}$ identity?

Answer

The identity only holds when the insurance and the annuity cover the exact same person, age, and time period. A term insurance and a whole life annuity are built from different random variables, so the shortcut does not apply.

#359 · Present Value Random Variables

What is the general formula for the variance of any random variable Z ?

Answer

$\text{Var}(Z) = E[Z^2] - (E[Z])^2$, the second moment minus the square of the mean.

#360 · Present Value Random Variables

What happens when you square $Z = v^{(K_x+1)}$?

Answer

$Z^2 = v^{2(K_x+1)} = (v^2)^{(K_x+1)}$, which has exactly the same mathematical form as Z but with v replaced by v^2 throughout.

#361 · Present Value Random Variables

What is 2A_x ?

Answer

2A_x is $E[Z^2]$, computed with the same formula as A_x but with every v replaced by v^2 (equivalently, using interest rate $(1+i)^2 - 1$). The mortality probabilities do not change.

#362 · Present Value Random Variables

Why is 2A_x called the result of 'double the force of interest'?

Answer

Because replacing v with v^2 is equivalent to valuing the insurance at the interest rate $(1+i)^2 - 1$, which corresponds to doubling the underlying force of interest used for discounting.

#363 · Present Value Random Variables

Write the formula for $\text{Var}(Z)$ using the double-force trick.

Answer

$$\text{Var}(Z) = \frac{1}{d^2} A_x - (A_x)^2.$$

#364 · Present Value Random Variables

How do you find $\text{Var}(Y)$ once you already know $\text{Var}(Z)$?

Answer

$\text{Var}(Y) = \text{Var}(Z) / d^2$, using the fact that $Y = (1-Z)/d$ is a linear transformation of Z , so no separate double-force annuity calculation is needed.

#365 · Present Value Random Variables

In the variance formula $\text{Var}(Y) = \text{Var}(Z)/d^2$, why does d get squared?

Answer

Variance scales by the square of any constant a random variable is multiplied by. Since Y is Z multiplied by the constant $-1/d$ (plus a shift), $\text{Var}(Y)$ equals $\text{Var}(Z)$ times $(1/d)^2$.

#366 · Present Value Random Variables

Given $l_x = 100,000$ and $l_{x+1} = 99,000$, find q_x .

Answer

$$q_x = (l_x - l_{x+1})/l_x = (100,000 - 99,000)/100,000 = 1,000/100,000 = 0.01.$$

#367 · Present Value Random Variables

Given $l_x = 100,000$ and $l_{x+3} = 95,000$, find ${}_3p_x$.

Answer

$${}_3p_x = l_{x+3} / l_x = 95,000 / 100,000 = 0.95.$$

#368 · Present Value Random Variables

With $i = 0.05$, what is v ?

Answer

$$v = 1/(1+i) = 1/1.05 = 0.952381.$$

#369 · Present Value Random Variables

With $i = 0.05$, what is d ?

Answer

$d = 1 - v = 1 - 0.952381 = 0.047619$ (equivalently $d = i/(1+i) = 0.05/1.05 = 0.047619$).

#370 · Present Value Random Variables

In the worked 3-year term insurance example ($i=0.05$, $q_x=0.01$, $q_{x+1}=0.0151515$, $q_{x+2}=0.0256410$), what is $A^1_{x:3}$?

Answer

$A^1_{x:3} = 0.0095238 + 0.0136054 + 0.0215959 = 0.044725$, summing v^{k+1} times $k p_x$ times q_{x+k} for $k = 0, 1, 2$.

#371 • Present Value Random Variables

In that same example, what is the pure endowment $A^{\{1\}}_{x:3}$?

Answer

$$A^{\{1\}}_{x:3} = v^3 \text{ times } {}_3p_x = 0.863838 \text{ times } 0.95 = 0.820646.$$

#372 • Present Value Random Variables

In that same example, what is the endowment insurance $A_{x:3}$?

Answer

$$A_{x:3} = A^1_{x:3} + A^{\{1\}}_{x:3} = 0.044725 + 0.820646 = 0.865371.$$

#373 · Present Value Random Variables

Using $A_{x:3} = 0.865371$ and $d = 0.047619$, find $\ddot{a}_{x:3}$.

Answer

$$\ddot{a}_{x:3} = (1 - A_{x:3})/d = (1 - 0.865371)/0.047619 = 0.134629/0.047619 = 2.8272.$$

#374 · Present Value Random Variables

In the worked example, what is ${}^2A^1_{x:3}$ for the term insurance (using v^2 , v^4 , v^6 in place of v , v^2 , v^3)?

Answer

$${}^2A^1_{x:3} = 0.0090703 + 0.0123405 + 0.0186554 = 0.0400662.$$

#375 · Present Value Random Variables

Using $A^1_{x:3} = 0.044725$ and ${}^2A^1_{x:3} = 0.0400662$, find $\text{Var}(Z)$ for the term insurance.

Answer

$$\text{Var}(Z) = {}^2A^1_{x:3} - (A^1_{x:3})^2 = 0.0400662 - 0.0020003 = 0.0380659.$$

#376 · Present Value Random Variables

What is the largest possible value Z can take, and when does it occur?

Answer

The largest value of Z is v^1 (the least amount of discounting), which occurs when $K_x = 0$, meaning death happens in the very first year.

#377 · Present Value Random Variables

As the year of death gets later and later, what happens to Z ?

Answer

Z gets smaller, approaching 0, since a larger exponent on v means more discounting and therefore a smaller present value.

#378 · Present Value Random Variables

What is the smallest possible value of the term insurance random variable for $A^1_{x:n}$?

Answer

0, which occurs whenever the insured survives past the end of the n -year term, since no death benefit is paid in that case.

#379 · Present Value Random Variables

Why is $A^{\{1\}}_{x:n}$ sometimes written as nEx ?

Answer

nEx is an alternative, shorter notation for the pure endowment, both meaning the actuarial present value of 1 paid only if the person age x survives n years.

#380 · Present Value Random Variables

What are the two separate 'worlds' combined inside any present value random variable?

Answer

A fixed, known interest world (v, i, d , set by the contract) and a random, unknown mortality world (K_x, k_{px}, q_{x+k} , determined by how long the person actually lives).

#381 • Present Value Random Variables

What is the domain weight of Present Value Random Variables on SOA Exam FAM?

Answer

This domain carries 15 percent of the total exam weight, tied for the largest single topic among the ten domains.

#382 • Present Value Random Variables

How many payments does an annuity-due make if the annuitant survives the entire n -year temporary period?

Answer

Exactly n payments, at times 0, 1, 2, up through $n-1$, since the annuity-due pays at the start of each year and stops once n years have been reached.

#383 · Present Value Random Variables

True or false: $\text{Var}(Z)$ can be computed using the same $k_p x$ and q_{x+k} values already used for A_x .

Answer

True. Only the discount factors change (v becomes v^2); the mortality probabilities $k_p x$ and q_{x+k} remain exactly the same in the 2A_x calculation.

#384 · Present Value Random Variables

Why is it incorrect to compute $\text{Var}(Z)$ as A_x squared minus A_x ?

Answer

Variance requires the second moment $E[Z^2]$, which is 2A_x , not A_x itself. The correct formula is $\text{Var}(Z) = {}^2A_x - (A_x)^2$, not any expression built from A_x alone.

#385 • Present Value Random Variables

What does the superscript 1 directly above the A signify in $A^1_{x:n}$?

Answer

It marks that the insurance benefit is contingent on death occurring within the stated period, distinguishing term insurance from a pure endowment.

#386 • Present Value Random Variables

What does the superscript 1 placed below and to the right signify in $A^{\{1\}}_{x:n}$?

Answer

It marks that the benefit is contingent on survival to the end of the stated period, identifying a pure endowment rather than a term insurance.

#387 · Present Value Random Variables

If $A_x = 0.4$ and $d = 0.05$, what is $\ddot{a}_x$?

Answer

$$\ddot{a}_x = (1 - A_x)/d = (1 - 0.4)/0.05 = 0.6/0.05 = 12.$$

#388 · Present Value Random Variables

If $\ddot{a}_x = 15$ and $d = 0.04$, what is A_x ?

Answer

$$A_x = 1 - d \text{ times } \ddot{a}_x = 1 - 0.04 \text{ times } 15 = 1 - 0.6 = 0.4.$$

#389 · Present Value Random Variables

What quantity must you find before you can compute either A_x or $\ddot{a}_x$ from a mortality table?

Answer

You must first find $k_p x$ and q_{x+k} (or equivalently the survival counts l_{x+k}) for every relevant future year, since both APVs are built from these probabilities.

#390 · Present Value Random Variables

What single random variable underlies both Z and Y ?

Answer

K_x , the curtate future lifetime. Both $Z = v^{(K_x+1)}$ and $Y = a - \text{angle} - (K_x+1)$ are functions of the exact same K_x .

#391 · Present Value Random Variables

How would you check a term insurance formula quickly using an extreme case?

Answer

Check what happens if death occurs in year one (present value close to the full benefit of 1) and what happens if the person survives the whole term (present value of 0), since term insurance only pays on death within the period.

#392 · Present Value Random Variables

How would you check a pure endowment formula quickly using an extreme case?

Answer

Check that immediate death pays nothing and that surviving exactly to time n pays the full discounted benefit v^n , since a pure endowment only pays on survival.

#393 · Present Value Random Variables

Why is $A_{x:n}$ always at least as large as $A^1_{x:n}$ alone?

Answer

Because $A_{x:n} = A^1_{x:n} + A^{\{1\}}_{x:n}$, and the pure endowment piece $A^{\{1\}}_{x:n}$ is always nonnegative, so adding it can only increase or leave unchanged the term insurance value.

#394 · Present Value Random Variables

What does 'discrete' mean in 'discrete whole life insurance'?

Answer

It means the death benefit is paid only at the end of the discrete policy year in which death occurs, rather than at the exact instant of death (which would be a continuous, fully continuous insurance model instead).

#395 · Present Value Random Variables

What is the annuity-due payment timing, and how does it differ from an annuity-immediate?

Answer

An annuity-due pays at the START of each year (times 0, 1, 2, ...), while an annuity-immediate pays at the END of each year (times 1, 2, 3, ...). The whole life annuity Y in this chapter is built on the annuity-due timing.

#396 · Present Value Random Variables

If $i = 0.05$, what interest rate corresponds to 'double the force of interest' for computing 2A_x ?

Answer

$(1+i)^2 - 1 = 1.05^2 - 1 = 1.1025 - 1 = 0.1025$, or 10.25 percent, since v^2 corresponds to discounting at this doubled rate.

#397 • Present Value Random Variables

Why does organizing k_{px} and q_{x+k} in a small table before multiplying by discount factors reduce exam errors?

Answer

It separates the mortality calculation from the interest calculation into two clean steps, which prevents the arithmetic slips that happen when both are combined in one long expression under time pressure.

#398 • Present Value Random Variables

What formula connects term insurance, pure endowment, and endowment insurance?

Answer

$A_{x:n} = A^1_{x:n} + A^{\{1\}}_{x:n}$. The endowment insurance is always the sum of the term insurance and the matching pure endowment on the same person and period.

#399 · Present Value Random Variables

In the deferred insurance formula, why does the sum start at $k = m$ instead of $k = 0$?

Answer

Because an m -year deferred insurance only pays a benefit if death occurs after the m -year waiting period has passed, so no earlier years of death (k less than m) contribute to the present value.

#400 · Present Value Random Variables

What are the two equivalent ways to write the actuarial present value of an m -year deferred whole life insurance?

Answer

Either sum over $k = m, m+1, \dots$ of v^{k+1} times $k p_x$ times q_{x+k} (the same term-by-term form as A_x , just starting later), or v^m times $m p_x$ times A_{x+m} (discount and survive to the deferral point, then apply an ordinary whole life insurance at the new age). Both give the identical value; they are not separate factors to combine.

#401 · Present Value Random Variables

Why is $\text{Var}(Z)$ always nonnegative?

Answer

Variance is defined as an average of squared deviations from the mean, and squared numbers can never be negative, so $\text{Var}(Z) = \frac{1}{n} \sum (A_x - \bar{A}_x)^2$ must always come out to zero or a positive number.

#402 · Present Value Random Variables

What is the practical use of computing $\text{Var}(Z)$ or $\text{Var}(Y)$ for an insurer?

Answer

The variance measures how spread out the present value outcomes are, which feeds directly into pricing risk margins, premium loadings, and capital requirements beyond just the expected cost captured by A_x or $\bar{a}_x$.

#403 · Present Value Random Variables

Is A_x the premium a policyholder pays for whole life insurance?

Answer

No. A_x is the actuarial present value of the benefit only, representing the expected cost to the insurer. The actual premium calculation, covered later in the syllabus, builds on A_x but also accounts for expenses and the timing of premium payments.

#404 · Present Value Random Variables

Summarize the single biggest shortcut formula from this chapter.

Answer

$A_x = 1 - d \text{ times } \ddot{a}_x$ (equivalently $\ddot{a}_x = (1 - A_x)/d$), which converts an insurance APV to its matching annuity APV, or the reverse, without resumming any mortality series.

#405 · Present Value Random Variables

If two people are the same age x with identical mortality, do they have the same Z ?

Answer

Not necessarily. Z depends on the actual realized value of K_x for each person, which can differ even between two people with identical mortality probabilities, since death timing is random for each individual.

#406 · Premium and Policy Value Calculation

State the equivalence principle used to calculate net premiums.

Answer

The actuarial present value of premiums equals the actuarial present value of benefits at issue:
 $EPV[\text{premiums}] = EPV[\text{benefits}]$. This balance is set at time zero, the moment the policy is issued.

#407 · Premium and Policy Value Calculation

What does EPV stand for and how does it differ from an ordinary present value?

Answer

EPV stands for expected present value, also called actuarial present value. Unlike an ordinary present value, it also weights each future cash flow by the probability that the triggering event (survival or death) actually occurs at that time.

#408 · Premium and Policy Value Calculation

What is a net premium?

Answer

A net premium is a premium calculated using only mortality and interest assumptions, with no allowance for the insurer's expenses. It is the theoretical premium found by applying the equivalence principle.

#409 · Premium and Policy Value Calculation

Write the formula for the annual net premium of a whole life insurance of 1, P_x .

Answer

$P_x = A_x / \ddot{a}_x$, where A_x is the EPV of the whole life benefit and $\ddot{a}_x$ is the EPV of a level annuity-due of 1 paid while the insured survives.

#410 · Premium and Policy Value Calculation

Why does the equivalence principle divide the benefit EPV by an annuity value to get the premium?

Answer

The annuity value $\ddot{a}_x$ represents the EPV of a stream of 1 per year while the insured survives. Dividing the total benefit EPV by this stream converts a single lump-sum present value into the level periodic premium that produces the same present value.

#411 · Premium and Policy Value Calculation

Write the net annual premium formula for an n-year term insurance.

Answer

$P = A^1_{x:n} / \ddot{a}_{x:n}$, where $A^1_{x:n}$ is the EPV of the death benefit if death occurs within n years, and $\ddot{a}_{x:n}$ is the EPV of premiums paid for at most n years.

#412 · Premium and Policy Value Calculation

Write the net annual premium formula for an n-year endowment insurance.

Answer

$P = A_{x:n} / \ddot{a}_{x:n}$, where $A_{x:n}$ is the EPV of the combined benefit paid on death within n years or on survival to the end of n years, whichever occurs first.

#413 · Premium and Policy Value Calculation

In endowment insurance notation, what is the difference between $A_{x:n}$ and $A^1_{x:n}$?

Answer

$A_{x:n}$ includes both the death benefit and the survival benefit (the endowment benefit). $A^1_{x:n}$ with the small 1 refers only to the death benefit portion, paid only if death occurs within the n years.

#414 · Premium and Policy Value Calculation

A whole life policy has $A_x = 0.25$ and $\ddot{a}_x = 12.5$ for a face amount of 1000. Find the annual net premium.

Answer

$P_x = A_x / \ddot{a}_x = 0.25 / 12.5 = 0.02$ per unit of face amount. For a face amount of 1000, the annual net premium is 1000 times 0.02, which equals 20.

#415 · Premium and Policy Value Calculation

Why is the term insurance annuity $\ddot{a}_{x:n}$ limited to n years even though the insured might live much longer?

Answer

Because premiums for a term policy stop being collected once the term ends. Even if the insured survives past year n , no further premiums are due, so the premium annuity only counts payments for the term period.

#416 · Premium and Policy Value Calculation

What is a gross premium?

Answer

A gross premium is the premium actually charged to a policyholder. Unlike a net premium, it also includes a loading to cover the insurer's expenses, such as commissions, underwriting, and administration.

#417 · Premium and Policy Value Calculation

Write the equivalence principle equation used to solve for a gross premium.

Answer

$EPV[\text{gross premiums}] = EPV[\text{benefits}] + EPV[\text{expenses}]$. The benefit side of the balance is expanded to include the present value of all expected future expenses.

#418 · Premium and Policy Value Calculation

Name the three common ways an expense can be modeled in a gross premium calculation.

Answer

A percent-of-premium expense, which scales with the premium charged, a per-policy expense, a flat amount regardless of policy size, and a per-1000-of-face expense, which scales with the death benefit amount.

#419 · Premium and Policy Value Calculation

What kind of insurer expense is usually modeled as a one-time cost at issue rather than a recurring cost?

Answer

Underwriting and policy issue costs are typically one-time expenses incurred only at the moment the policy is issued, unlike per-policy administration expenses which recur every year the policy stays in force.

#420 · Premium and Policy Value Calculation

A whole life policy (face 1000, $A_x = 0.25$, $\ddot{a}_x = 12.5$) has expenses of 10 percent of premium, 2 per year flat, and a one-time issue expense of 25. Set up the gross premium equation.

Answer

$G * \ddot{a}_x = (1000 * A_x) + (0.10 * G * \ddot{a}_x) + (2 * \ddot{a}_x) + 25$, which becomes $G * 12.5 = 250 + 1.25G + 25 + 25$.

#421 · Premium and Policy Value Calculation

Solve for G in: $G * 12.5 = 250 + 1.25G + 25 + 25$.

Answer

Collect the G terms: $G*(12.5 \text{ minus } 1.25) = 300$, so $G*11.25 = 300$, giving $G = 300/11.25 = 26.67$.

#422 · Premium and Policy Value Calculation

In the gross premium example where the net premium was 20 and the gross premium was 26.67, what explains the difference?

Answer

The difference of 6.67 is entirely due to the expense loading: the percent-of-premium charge, the flat per-policy expense, and the one-time issue expense, none of which appear in the net premium calculation.

#423 · Premium and Policy Value Calculation

Why does a percent-of-premium expense make the gross premium equation harder to solve directly than a flat expense?

Answer

Because the unknown gross premium G appears on both sides of the equation (once as the premium itself and once inside the expense term), so the G terms must be collected and combined before solving, rather than solving in one step.

#424 · Premium and Policy Value Calculation

What is a policy value, and what was it historically called?

Answer

A policy value, historically called a reserve, is the amount of money an insurer needs to hold at a given point after issue so it can meet its future obligations on that policy.

#425 · Premium and Policy Value Calculation

Why is a policy value needed even though the policyholder pays a level premium every year?

Answer

Because the true cost of insurance rises every year as the insured ages, while the premium stays level. Early premiums collect more than that year's true cost and build up a fund; later premiums collect less, and the fund built up earlier is drawn down to cover the shortfall.

#426 · Premium and Policy Value Calculation

What is the policy value at time zero, immediately at issue, and why?

Answer

The policy value at time zero is zero. This follows directly from the equivalence principle, since the EPV of premiums exactly equals the EPV of benefits at issue, leaving nothing left over.

#427 · Premium and Policy Value Calculation

What does the notation tV_x mean?

Answer

tV_x is the policy value at duration t for a policy issued to a life aged x , representing the fund the insurer holds on that policy at that point in time.

#428 · Premium and Policy Value Calculation

Describe the prospective method for calculating a policy value in plain words.

Answer

The prospective method looks forward from the valuation date and asks what the insurer still expects to pay out in future benefits, minus what it still expects to collect in future premiums, given the policy is in force.

#429 · Premium and Policy Value Calculation

Write the prospective policy value formula for whole life insurance at duration t .

Answer

${}^tV_x = A_{x+t}$ minus $(P_x * \ddot{a}_{x+t})$, where both A_{x+t} and $\ddot{a}_{x+t}$ are valued at the insured's attained age of $x+t$.

#430 · Premium and Policy Value Calculation

In the prospective policy value formula, why is the attained age $x+t$ used instead of the original issue age x ?

Answer

Because the calculation is being done at duration t , and the future benefits and premiums must be valued based on the insured's current age at that time, which is $x+t$, not the age at original issue.

#431 · Premium and Policy Value Calculation

Describe the retrospective method for calculating a policy value in plain words.

Answer

The retrospective method looks backward and asks how much money has accumulated so far, from past premiums collected minus past benefits and costs paid out, carried forward with interest, per policy still in force.

#432 · Premium and Policy Value Calculation

Why must the prospective and retrospective policy values be equal at the same duration?

Answer

Because the equivalence principle guarantees that premiums and benefits balance exactly over the whole lifetime of the policy, so the same underlying fund must be described whether it is measured by looking forward or by looking backward, as long as the same assumptions are used both ways.

#433 · Premium and Policy Value Calculation

In a retrospective calculation, why is the accumulated fund divided by the probability of survival to that duration?

Answer

The accumulated fund is first computed per original policy issued. Dividing by the survival probability re-expresses that fund per policy still in force, since only survivors are holding a policy value at that point.

#434 · Premium and Policy Value Calculation

A 3-year term insurance of 1000 has $q_x = 0.01$, $i = 5$ percent, and a net annual premium of 11.35. Find the retrospective fund per original policy after year 1, before dividing by survivorship.

Answer

Accumulate the premium with interest: $11.35 * 1.05 = 11.92$. Subtract the expected cost of claims in year 1: $0.01 * 1000 = 10$. The fund per original policy is 11.92 minus 10 = 1.92.

#435 · Premium and Policy Value Calculation

Continuing the prior example, with $p_x = 0.99$, find the retrospective policy value per surviving policy at duration 1.

Answer

Divide the accumulated fund of 1.92 by the survival probability 0.99: $1.92 / 0.99 = 1.94$, which matches the prospective value at duration 1 within rounding.

#436 · Premium and Policy Value Calculation

For the same 3-year term insurance, the prospective future benefit EPV at duration 1 is 23.98 and the future premium EPV is 22.03. Find the prospective policy value at duration 1.

Answer

$1V = 23.98$ minus $22.03 = 1.95$, which matches the retrospective value of 1.94 within ordinary rounding tolerance.

#437 · Premium and Policy Value Calculation

What assumptions must match for the prospective and retrospective policy values to agree exactly?

Answer

Both methods must use the same mortality assumptions, the same interest assumption, and the same premium, since any difference in assumptions between the two views would break the balance guaranteed by the equivalence principle.

#438 · Premium and Policy Value Calculation

Write the general recursive relation for policy values from duration t to duration $t+1$.

Answer

$(tV + P) * (1+i) = (q_{x+t} * \text{benefit}) + (p_{x+t} * (t+1)V)$, meaning the policy value plus the premium, accumulated with one year of interest, funds the expected claim cost plus the expected policy value carried forward for survivors.

#439 · Premium and Policy Value Calculation

Rearrange the recursive relation to solve for the policy value one year later, $(t+1)V$.

Answer

$$(t+1)V = [(tV + P) * (1+i) \text{ minus } (q_{\{x+t\}} * \text{benefit})] / p_{\{x+t\}}$$

#440 · Premium and Policy Value Calculation

In the recursive formula, why is the sum $(tV + P)$ accumulated with interest before subtracting the cost of claims?

Answer

Because the existing policy value and the new premium are both available at the start of the year and earn one full year of interest together before any claims are paid at the end of the year, so both amounts must be added and accumulated together first.

#441 · Premium and Policy Value Calculation

For a 3-year term insurance with $1V = 1.94$, $P = 11.35$, $i = 5$ percent, $q_{x+1} = 0.012$, and $p_{x+1} = 0.988$, find $2V$ using the recursion.

Answer

Accumulate: $(1.94 + 11.35) \cdot 1.05 = 13.9545$. Subtract the claim cost: 13.9545 minus $(0.012 \cdot 1000) = 1.9545$. Divide by survival probability: $1.9545 / 0.988 = 1.98$.

#442 · Premium and Policy Value Calculation

Why is the recursive relation often faster than the prospective method on an exam question?

Answer

Because if a policy value at one duration is already known, the recursion moves it forward by one year using only the mortality, interest, and premium for that single year, avoiding the need to recompute a full multi-year prospective actuarial present value from scratch.

#443 · Premium and Policy Value Calculation

For a term insurance, what must the policy value equal at the exact end of the term, and why?

Answer

The policy value must equal zero at the end of the term, since the coverage ends with no further benefit possible and no further premiums due, so nothing remains to be held.

#444 · Premium and Policy Value Calculation

What is the net amount at risk in the context of a policy value recursion?

Answer

The net amount at risk is the difference between the death benefit and the policy value already held (benefit minus tv). It represents the true fresh loss to the insurer if death occurs, since the policy value itself would already have been available.

#445 · Premium and Policy Value Calculation

What is Thiele's differential equation the continuous-time analogue of?

Answer

Thiele's differential equation is the continuous-time analogue of the discrete recursive relation for policy values, used when premiums are paid continuously and mortality is described by a force of mortality rather than an annual probability.

#446 · Premium and Policy Value Calculation

Write Thiele's differential equation for a policy with continuous premium rate $P(t)$ and death benefit $S(t)$.

Answer

$$\frac{d}{dt}[tV] = (\delta * tV) + P(t) \text{ minus } [\mu_{\{x+t\}} * (S(t) \text{ minus } tV)],$$
 where δ is the force of interest and $\mu_{\{x+t\}}$ is the force of mortality at attained age $x+t$.

#447 · Premium and Policy Value Calculation

In Thiele's differential equation, what does the term δtV represent?

Answer

It represents the interest that the existing policy value earns continuously, where δ is the force of interest, the continuous-time analogue of the annual interest rate i .

#448 · Premium and Policy Value Calculation

In Thiele's differential equation, what does the term $P(t)$ represent?

Answer

$P(t)$ is the continuous rate at which premium income is flowing into the policy value fund at time t .

#449 · Premium and Policy Value Calculation

In Thiele's differential equation, what does μ_{x+t} times $(S(t) \text{ minus } tV)$ represent?

Answer

It represents the expected instantaneous cost of the mortality risk the insurer is still carrying, equal to the force of mortality times the net amount at risk, which is the death benefit minus the policy value already held.

#450 · Premium and Policy Value Calculation

In plain words, what does Thiele's differential equation say about how the policy value changes over time?

Answer

The policy value grows through interest earned and premium income coming in, and shrinks through the expected cost of the mortality risk that is not already covered by the fund already on hand.

#451 · Premium and Policy Value Calculation

Why is Thiele's differential equation usually tested at a conceptual level rather than through numeric integration on the exam?

Answer

Because understanding what each term in the equation represents in words (interest earned, premium income, and the cost of the uncovered mortality risk) is generally more valuable for answering exam questions than performing the calculus needed to solve the differential equation directly.

#452 · Premium and Policy Value Calculation

What is the net amount at risk used for in Thiele's equation, and why does it use $(S(t) - tV)$ instead of just $S(t)$?

Answer

It measures the true fresh loss to the insurer on death, because the policy value tV would already have been available as a fund to help pay the claim, so only the excess of the benefit over that fund represents new risk to the insurer.

#453 · Premium and Policy Value Calculation

What is the single most common exam trap when applying the recursive policy value formula?

Answer

Mixing up the direction of the recursion, especially forgetting that the existing policy value and the premium just paid must both be included and accumulated with one year of interest before subtracting the expected claim cost.

#454 · Premium and Policy Value Calculation

What is the single most common exam trap when choosing between a net premium formula and a gross premium formula?

Answer

Applying a net premium formula to a question that clearly describes expense-loaded, gross premiums, or the reverse; carefully identifying which cash flows belong on which side of the equivalence principle avoids this error.

#455 · Premium and Policy Value Calculation

Why is Premium and Policy Value Calculation typically the highest-weighted single topic on the FAM exam?

Answer

Because nearly every calculation-heavy question on the exam reduces to setting up the equivalence principle, computing a prospective or retrospective policy value, or stepping a policy value forward with the recursive relation, making it the most heavily and consistently tested domain.

#456 · Premium and Policy Value Calculation

For an n-year endowment insurance, why does the annuity in the premium formula run only for the term of the policy, $\ddot{a}_{x:n}$, rather than for life?

Answer

Because the endowment policy itself only lasts n years, ending either in death or survival to duration n, so no premiums are ever due beyond that point regardless of the annuity's underlying life table.

#457 · Premium and Policy Value Calculation

If A_x increases while $\ddot{a}_x$ stays the same, what happens to the net annual premium P_x , and why?

Answer

P_x increases, because $P_x = A_x / \ddot{a}_x$, so a larger EPV of benefits with the same premium annuity value requires a larger annual premium to keep the equivalence principle balanced.

#458 · Premium and Policy Value Calculation

If $\ddot{a}_x$ increases while A_x stays the same, what happens to the net annual premium P_x , and why?

Answer

P_x decreases, because the same total benefit EPV is now being spread over a larger annuity value, meaning a smaller level premium is enough to accumulate the same present value.

#459 · Premium and Policy Value Calculation

What does it mean, in terms of the equivalence principle, for a premium to be described as 'expense-loaded'?

Answer

It means the premium has been calculated so that its EPV covers not just the EPV of the insurance benefit, but also the EPV of the insurer's expenses, as opposed to a net premium which covers only the benefit.

#460 · Premium and Policy Value Calculation

In the gross premium worked example, what would happen to the gross premium if the one-time issue expense of 25 were removed but everything else stayed the same?

Answer

The equation would become $G \cdot 12.5 = 250 + 1.25G + 25$, so $G \cdot 11.25 = 275$, giving $G = 275 / 11.25 = 24.44$, a lower gross premium than the 26.67 found with the issue expense included.

#461 · Premium and Policy Value Calculation

Why does a per-policy expense affect a small policy proportionally more than a large policy?

Answer

Because a per-policy expense is a flat dollar amount regardless of face amount, so for a small face amount it represents a larger share of the total premium than it does for a large face amount, where the same flat cost is spread over more coverage.

#462 · Premium and Policy Value Calculation

What is the purpose of setting the policy value at issue equal to zero when deriving the recursive formula for the very first year?

Answer

Setting OV equal to zero reflects that the equivalence principle already balanced premiums and benefits exactly at issue, so there is no leftover fund, and the recursion for the first policy year begins from that zero starting point.

#463 · Premium and Policy Value Calculation

Explain in one sentence why the prospective policy value formula subtracts a premium annuity term rather than adding it.

Answer

Because future premiums are money the insurer still expects to receive and can use to offset future benefit costs, so they reduce the amount of fund the insurer needs to be holding right now.

#464 · Premium and Policy Value Calculation

What actuarial function represents the EPV of a level annuity-due of 1 paid annually while a life aged x survives?

Answer

$\ddot{a}_x$, read as 'a-double-dot sub x', is the actuarial present value of an annuity-due of 1 per year paid at the start of each year while the insured is alive.

#465 · Premium and Policy Value Calculation

What actuarial function represents the EPV of a whole life insurance benefit of 1 on a life aged x ?

Answer

A_x is the actuarial present value of a whole life insurance of 1, payable whenever the insured eventually dies, however far in the future that occurs.

#466 · Premium and Policy Value Calculation

In the notation $A^1_{x:n}$, what does the superscript 1 above the A indicate?

Answer

It indicates that the benefit is a death benefit contingent on death occurring within the n -year term, as opposed to a survival or endowment benefit.

#467 · Premium and Policy Value Calculation

Describe, in one sentence, the balance-scale picture of the equivalence principle.

Answer

One side of the scale holds all future premium payments weighted by survival probability and discounted to today, the other side holds all future benefit payments weighted by the probability of the triggering event and discounted to today, and the equivalence principle sets these two sides equal at issue.

#468 · Premium and Policy Value Calculation

Why is a net premium sometimes called an 'equivalence premium'?

Answer

Because it is the premium that results directly from solving the equivalence principle equation, $EPV[\text{premiums}] = EPV[\text{benefits}]$, with no additional loading for expenses.

#469 · Premium and Policy Value Calculation

For a 3-year term insurance, why is the EPV of the death benefit built from three separate terms, one for each policy year?

Answer

Because death can occur in year 1, year 2, or year 3, and each possibility requires its own discount factor and its own probability of surviving to the start of that year and then dying during it, so each year's contribution must be calculated and summed separately.

#470 · Premium and Policy Value Calculation

In the 3-year term insurance example, why does the term for death in year 3 include both p_x and p_{x+1} in its probability factor?

Answer

Because to die during year 3, the insured must first survive year 1 (probability p_x) and survive year 2 (probability p_{x+1}), and only then die during year 3, so both survival probabilities must be multiplied together with the year-3 mortality rate.

#471 · Premium and Policy Value Calculation

What role does the discount factor v play in each term of a death benefit EPV calculation?

Answer

The discount factor v , raised to the power of the number of years until the benefit could be paid, converts a benefit paid at a future point in time back into its equivalent value in today's dollars.

#472 · Premium and Policy Value Calculation

Why must a policy value calculation for a term insurance eventually reach zero, while a whole life policy value never reaches zero while the insured is alive?

Answer

A term insurance ends after a fixed number of years with no further obligation, so its policy value winds down to zero by the end of the term, while a whole life policy continues indefinitely as long as the insured is alive, so its policy value does not have a natural endpoint before death.

#473 · Premium and Policy Value Calculation

If a recursive policy value calculation produces a negative policy value, what should that suggest to the solver?

Answer

It should prompt a check of the inputs and arithmetic, since for a standard level-premium policy without unusual loading, the policy value is not normally expected to become negative; a negative result often signals an error in setting up the recursion or in the given assumptions.

#474 · Premium and Policy Value Calculation

In the equivalence principle equation for a gross premium, why does the expense term for a percent-of-premium charge use G rather than the net premium P ?

Answer

Because a percent-of-premium expense is defined as a percentage of whatever premium the policyholder actually pays, which is the gross premium G , not the theoretical net premium P that ignores expenses.

#475 · Premium and Policy Value Calculation

What is the practical business reason an insurer needs to hold a policy value rather than simply spending each year's premium as it is collected?

Answer

Because premiums are level while the true cost of insurance rises with age, so the insurer must set aside the excess collected in early years in order to have enough funds available to pay the larger expected claim costs in later years.

#476 · Premium and Policy Value Calculation

Fill in the blank: the policy value 'building up' picture shows the prospective view starting from the _____ and moving backward in time to meet the retrospective view.

Answer

future (or: end of the policy). The prospective view calculates the fund's expected size by working from future benefits and premiums backward to the valuation date, meeting the retrospective view, which works forward from issue, at the same point.

#477 · Premium and Policy Value Calculation

Why is it important that the recursive formula uses $p_{\{x+t\}}$ and $q_{\{x+t\}}$, the mortality rates at the CURRENT attained age, rather than the original issue-age rates?

Answer

Because the recursion is stepping the policy value forward one specific policy year at a time, and the correct mortality experience for that year is based on how old the insured actually is at that point, which is the attained age $x+t$, not the original age at issue.

#478 · Premium and Policy Value Calculation

In the equivalence principle, if benefits are paid at the end of the year of death, what timing convention should be used for the discount factor applied to each year's death benefit?

Answer

The death benefit for deaths in year k should be discounted by v^k , since the benefit is not paid until the end of the k -th policy year, even though death itself may occur at any point during that year.

#479 · Premium and Policy Value Calculation

What is the effect on the net premium if a whole life policy is instead written as an n-year term policy with the same mortality and interest assumptions, all else equal?

Answer

The term insurance premium is generally lower per unit of coverage during the term than the level whole life premium, because the term policy only pays a benefit if death occurs within the limited period and does not need to fund the higher costs of insurance at very advanced ages.

#480 · Premium and Policy Value Calculation

In a policy value recursion, what does it mean if p_{x+t} appears in the denominator when solving for $(t+1)V$?

Answer

It means the calculated cash flow is being spread only over the policies that actually survive to duration $t+1$, since only survivors will be holding that next policy value, so dividing by the survival probability correctly reallocates the amount per surviving policyholder.

#481 · Premium and Policy Value Calculation

Why does an endowment insurance typically have a much larger net premium than an otherwise identical term insurance?

Answer

Because an endowment insurance guarantees a payout even if the insured survives to the end of the term, adding a savings-like survival benefit on top of the death benefit, which substantially raises the EPV of benefits that the premium must fund.

#482 · Premium and Policy Value Calculation

What two actuarial present values must be computed before applying the formula $P_x = A_x / \ddot{a}_x$?

Answer

The EPV of the whole life insurance benefit, A_x , and the EPV of the level premium annuity-due, $\ddot{a}_x$, both of which are typically taken from or derived using a life table and an assumed interest rate.

#483 · Premium and Policy Value Calculation

Why is the equivalence principle applied only ONCE, at the moment of issue, rather than being re-applied every policy year?

Answer

Because the level premium and the benefit structure are fixed at issue based on the balance achieved at that single point in time; the policy value at later durations then measures how that original balance is playing out, rather than resetting the balance each year.

#484 · Premium and Policy Value Calculation

What is the plain-English meaning of the phrase 'expected present value' as used throughout this chapter?

Answer

It means the present value of a future cash flow after being multiplied by the probability that the event triggering that cash flow (such as survival to a certain year or death in a certain year) actually happens.

#485 · Premium and Policy Value Calculation

In Thiele's differential equation, why is the term for mortality cost subtracted rather than added?

Answer

Because the expected cost of the mortality risk represents an outflow that reduces the policy value fund, offsetting the inflows from interest earned and premium income, so it must be subtracted to correctly describe the net rate of change of the fund.

#486 · Premium and Policy Value Calculation

How does the concept of a policy value connect the equivalence principle at issue to the ongoing life of a policy?

Answer

The equivalence principle guarantees the premiums and benefits balance exactly at issue with a starting policy value of zero; the policy value at every later duration then tracks how that original balance has evolved as some policyholders have died, some have survived, and interest has accrued.

#487 · Premium and Policy Value Calculation

What common exam phrasing signals that a question wants the retrospective method rather than the prospective method?

Answer

Phrasing that describes what has already happened on the policy, such as premiums already collected or claims already paid up to the current duration, signals a retrospective calculation, as opposed to phrasing describing what remains to happen in the future, which signals a prospective calculation.

#488 · Premium and Policy Value Calculation

Why can the recursive relation be used to check a prospective or retrospective calculation for consistency?

Answer

Because if the policy value at one duration and the assumptions for the following year are known, the recursive formula predicts the next duration's policy value; if this predicted value does not match a value found by prospective or retrospective calculation, it signals an arithmetic or assumption error somewhere.

#489 · Premium and Policy Value Calculation

What is the net amount at risk formula, and where does it appear outside of Thiele's equation?

Answer

The net amount at risk equals the death benefit minus the policy value already held (benefit minus tV). The same quantity, in discrete form, is what makes each year's expected claim cost only partially offset the fund already being carried forward for survivors in the yearly recursive relation.

#490 · Premium and Policy Value Calculation

Why is it necessary to state both an interest assumption and a mortality assumption before any net premium or policy value can be calculated?

Answer

Because every actuarial present value combines a discounting effect from interest and a probability-weighting effect from mortality; without both assumptions specified, neither the benefit side nor the premium side of the equivalence principle can be evaluated.

#491 · Premium and Policy Value Calculation

In the worked 3-year term insurance example, why does the annuity $\ddot{a}$ (for premiums) have one fewer year of mortality decrement built in than the full 3-year insurance benefit?

Answer

Because the annuity pays at the START of each year while the policy is in force, so the third payment (at the start of year 3) only requires survival through the first two years, whereas the insurance benefit calculation must account for death occurring in any of all three years, including the third.

#492 · Premium and Policy Value Calculation

What is the key difference between how 'per-policy' and 'per-1000-of-face' expenses scale with policy size?

Answer

A per-policy expense is a fixed dollar amount that does not change no matter how large or small the face amount is, while a per-1000-of-face expense increases proportionally as the face amount increases, since it is charged per unit of coverage rather than per policy.

#493 · Premium and Policy Value Calculation

Why would an actuary want to verify a computed policy value using BOTH the prospective and retrospective methods rather than only one?

Answer

Computing both provides an independent cross-check: if the equivalence principle assumptions were applied consistently and correctly, the two methods must produce the same answer, so agreement between them confirms the calculation, while a mismatch reveals an error.

#494 · Premium and Policy Value Calculation

What does the force of interest, delta, represent in Thiele's differential equation, and how does it relate to the annual interest rate i used elsewhere in this chapter?

Answer

The force of interest δ is the continuous, instantaneous rate at which money grows, used in continuous-time models like Thiele's equation, while the annual interest rate i is the discrete, once-per-year growth rate used in the recursive formula and annuity calculations; they describe the same underlying interest earning process at different time granularities.

#495 · Premium and Policy Value Calculation

Summarize, in one sentence, what makes the equivalence principle 'work' as a fair basis for setting a premium.

Answer

It works because it sets the premium at exactly the level where the value the policyholder expects to pay in, given realistic survival probabilities, equals the value the insurer expects to pay out, given the same probabilities, leaving no built-in advantage to either side before any expenses are considered.

#496 · Premium and Policy Value Calculation

For a whole life policy, does the policy value tV_x generally increase, decrease, or stay level as duration t increases toward very old age?

Answer

The policy value generally increases toward the face amount as duration increases, since the insured is aging and the probability of imminent death is rising, so the fund needed to cover the now-nearer expected claim grows steadily larger.

#497 · Premium and Policy Value Calculation

In a recursive policy value step, what happens to $(t+1)V$ if the assumed mortality rate q_{x+t} is understated?

Answer

If q_{x+t} is understated, too little is subtracted for the expected claim cost, which makes the computed $(t+1)V$ come out too high relative to the value that would result from the correct mortality rate.

#498 · Premium and Policy Value Calculation

Why does the gross premium equation in this chapter place the one-time issue expense of 25 outside of any multiplication by $\ddot{a}_x$?

Answer

Because the issue expense is a single cost incurred only once, at time zero when the policy begins, rather than a recurring annual cost, so it is added directly rather than being multiplied by an annuity value that spreads a cost over multiple years.

#499 · Premium and Policy Value Calculation

What is the relationship between the net premium P_x and the pure cost of insurance in the very first policy year, in plain terms?

Answer

In the first year the level net premium P_x typically exceeds the pure cost of insuring the current year's mortality risk, and this excess is what begins building the policy value fund that will be needed to cover the higher costs of insurance in later years.

#500 · Premium and Policy Value Calculation

Why is it misleading to think of a policy value as money the policyholder personally 'owns' the way a bank savings balance would work?

Answer

Because the policy value is a pooled actuarial fund calculated across a whole cohort of similar policies using survival probabilities; it represents the amount the insurer needs to be holding on average per in-force policy, not a literal individual savings account belonging to one specific policyholder.